Guidelines for the use of AI in language education

The AI for Language Education group

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Executive summary

As artificial intelligence (AI) transforms the way that languages are taught and learned, the *Guidelines for the use of AI in language education* aim to provide language teachers, school leadership and other policy makers and stakeholders with support and guidance so that they can make ethical, pedagogically appropriate and confident decisions in their day-to-day practice. Developed under the AI Lang project of the European Centre for Modern Languages, this document provides readers with clear principles, guidelines, competence descriptors, illustrative examples and reflective tools that connect sound pedagogical practice with the affordances of emerging technologies. This guidance is rooted in European values of human rights, democracy and respect for linguistic diversity, and it aims to serve both as a scaffold for practice and as a catalyst for professional growth in an AI-enhanced educational setting.

Background

The first chapter of the *Guidelines* aims to contextualise the guidance that will follow. It presents a broad overview of how artificial intelligence is reshaping language education, with particular reference to the affordances and challenges that are specific to language teaching and learning. It also includes a typology of artificial intelligence systems that are used to support language education. This chapter also explores existing guidance on the use of AI, with a view to highlighting shared concerns and identifying questions and challenges that remain unresolved.

Guidelines

The second chapter presents a framework of guidelines for use by language teachers. These are prefaced by two brief sections outlining the scope and structure of the framework as well as a set of principles on which the framework is grounded, namely the **safe**, **responsible, purposeful** and **human-centred** use of artificial intelligence. Each principle is then operationalised in two specific guidelines, which are further translated into competences. To facilitate engagement with each guideline, readers are presented with reflection questions that encourage them to identify their strengths and areas for growth.

Usage cases

In the third chapter, readers are presented with six cases of AI use in various contexts across Europe. These cases highlight both the diversity of language education approaches and the many ways in which AI is being used. Each case is accompanied by commentary that builds on the guidance presented in this document. The cases exemplify how the framework can be adapted across different educational realities, and they are designed to help readers consider how AI-enhanced language education can be responsibly and effectively implemented irrespective of institutional scale and availability of resources.

Glossary

The document also includes a section listing key terms for the readers' reference. Terms that appear in the glossary are highlighted in the text.

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Preface

Many professionals in language education recall precisely where they were in late 2022 when the first publicly accessible large language model chat interfaces appeared. Artificial intelligence (AI) itself was not a new concept, yet the unprecedented ease with which individuals worldwide could explore and harness this transformative technology marked a turning point. In the years since, this new form of human-machine interaction and its rapid pace of adoption continue to unsettle established practices across society, including in language education. While innovations such as AI-powered intelligent tutoring systems hold promise, AI also raises significant ethical, privacy and data governance concerns, and there is a pressing need for clear, ethically grounded policy frameworks and pedagogical guidance to bridge the rapidly evolving field of AI with the practical realities of language classrooms.

The AI Lang project seeks to respond to this need by offering a conceptual framework with shared definitions, typologies of artificial intelligence and functional distinctions, and providing a common AI vocabulary for the language education community. The framework outlines core principles for AI-assisted language teaching and learning, and provides resources including guidelines, competence descriptors, reflection tasks, and usage cases to illustrate ethical implementation.

Although this guidance has been produced within a European institutional context, the concerns it addresses extend beyond this context. Similar publications are being produced elsewhere, including national AI strategies and education-sector guidance across the Americas, Asia and Africa, alongside guidelines developed by professional associations. Readers are encouraged to read the guidance that follows alongside the regulatory and policy instruments that apply in their own jurisdiction.

Our primary audience includes language educators, but the guidance is also relevant to teacher trainers, administrators and policy authors. The ultimate beneficiaries of the guidance, of course, are language learners, and especially those who might be rendered vulnerable by the uncritical adoption of AI-supported teaching practices. By foregrounding the competences required for informed tool use (AI literacy), the document situates language education within broader conversations across the social sciences, humanities, and STEM.

At the same time, the framework deliberately centres on the unique role of language and language education, which is being reshaped under the influence of artificial intelligence. Rather than being one more subject among many, language is the medium through which all learning takes place. As a result, artificial intelligence tools that mediate language learning shape how learners engage with knowledge. Still more importantly, language education bears a special responsibility for fostering effective communication skills, critical reading and argumentation, which form the foundations of democratic citizenship. Finally, set against a backdrop of algorithmically generated content, language education can leverage a long tradition of attending to meaning, context, identity and power, fostering critical understanding and the ability to produce language with intention.

In this guidance, we view artificial intelligence as a sociocultural construct, intertwined with questions of equity, environmental sustainability, and language policy. Central to

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the discussion is a firm commitment to human agency: teachers and learners must remain the primary architects of learning processes, ensuring that technology enhances rather than replaces expertise. A social justice perspective further examines algorithmic bias, datafication, and digital divides, while advancing strategies to position AI as a means of inclusion rather than exclusion.

The document comprises five parts. The Background outlines the potential of AI for language education, defines key terms and typologies, and discusses the concerns about the optimal use of the emerging affordances. The Guidelines present four core principles and eight guidelines, each with competence descriptors and reflection prompts. The usage cases illustrate these guidelines in practice. Finally, the Glossary and References provide essential terminology and links to the supporting research and frameworks underpinning this report.

Collaboratively authored by scholars from across Europe and Canada, the framework draws on peer-reviewed research and key policy documents from leading organisations such as UNESCO, the OECD and the Council of Europe. It also incorporates practices piloted in various educational settings, with iterative feedback loops to validate and refine the recommendations presented. We hope these guidelines can serve as a foundation for responsible experimentation and collective reflection on how to harness the potential of AI in ways that strengthen human oversight and advance plurilingual and pluricultural language education for all.

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Acknowledgment

This document is one of the main outcomes of the Artificial Intelligence for Language Education (AI Lang) project, which ran from 2024 to 2027 with the support of the European Centre for Modern Languages.

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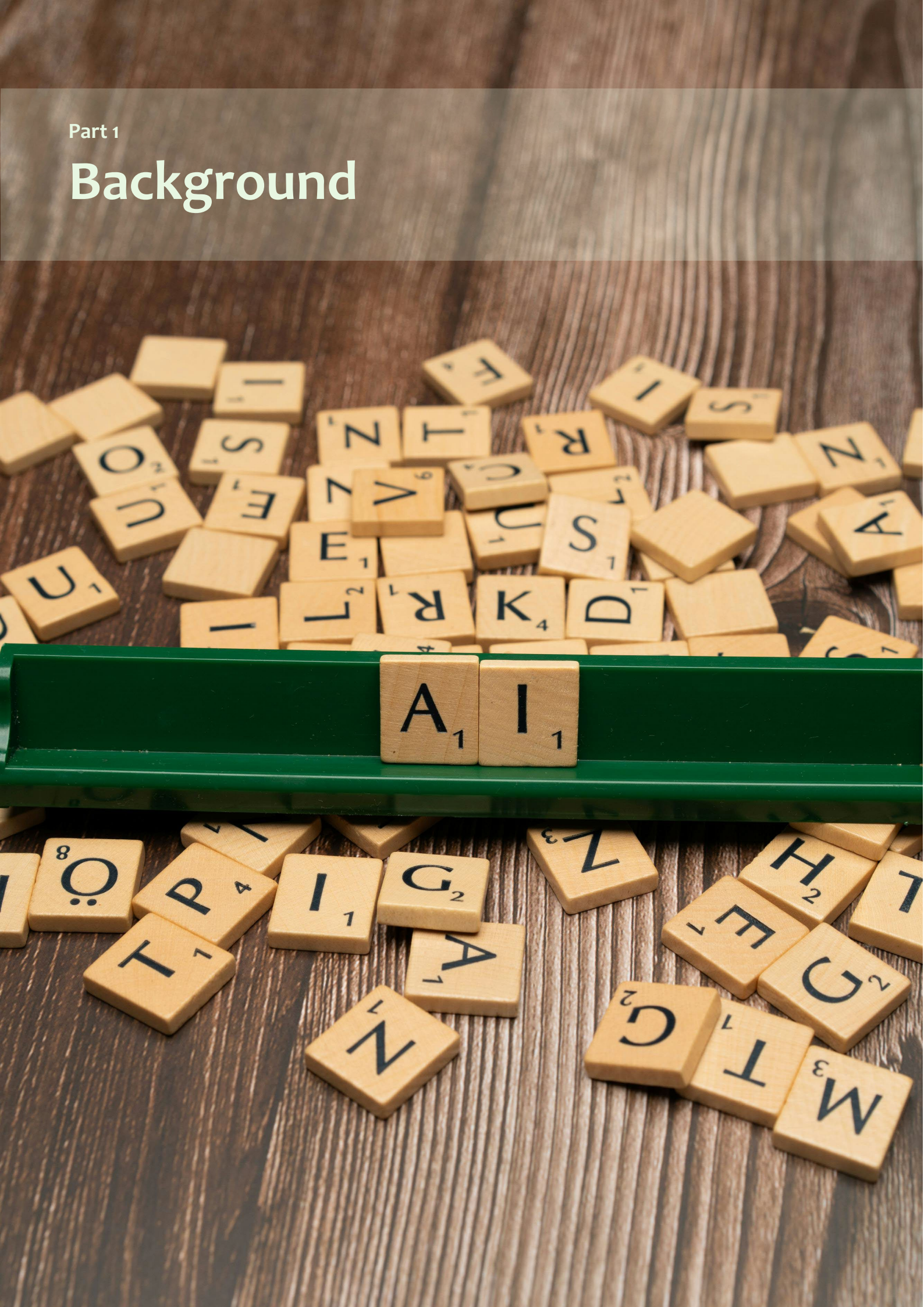
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The authors remain responsible for any omissions, errors and other infelicities in the text.

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Part 1

Background



1. Background

I believe that at the end of the century [...] *one will be able to speak of machines thinking without expecting to be contradicted.*
Alan Turing (1950, §6, p. 442)

The rapid rise of artificial intelligence (AI), supported by advancements in **big data** and cybersecurity, is driving transformative and sometimes controversial changes across professional and social life.

These changes are being felt in the world of education, where the increasing integration of AI-powered digital tools is reshaping how languages are taught and learned. According to the *Future of Jobs Report 2025* (Leopold et al., 2025), 60% of employers believe that by 2030, the labour market will have been transformed by expanding digital access. In the same report, it is predicted that nearly 40% of the skills people use today will need to be updated or replaced in the same timeframe. While these predictions suggest that the demand for teaching and learning is likely to see significant growth, the nature of teaching work is also expected to see extensive change as AI becomes embedded in educational systems. For this reason, developing AI-related professional competence, including critical AI literacy, ethical judgment and the ability to design, supervise and assess AI-mediated learning, is increasingly becoming a core requirement for effective and professionally relevant practice.

At the same time, it is essential that AI-assisted teaching and learning serves, rather than shapes, the goals of language education, and that it does so by aligning with core pedagogical principles and values, as well as local needs and priorities. The challenge, in other words, is not merely to adopt AI, but to do so in ways that promote equity, enhance agency and remain sensitive to the specific needs of diverse educational contexts. As technological advances outpace our ability to effectively theorise, identify possibilities, and reflect on their broader implications for language education, a survey of the field seems timely and needed to establish a theoretical backdrop for the guidelines that will follow.

This brief review first examines the potential of artificial intelligence in language education (Section 1.1). In the following section (Section 1.2), key terms are defined, with particular reference to language teaching and learning and the different typologies that can be used to conceptually map the field. Finally, Section 1.3 surveys relevant guidance for language teachers, synthesises areas of shared ground, and identifies questions and challenges that remain to be addressed.

1.1. Situating AI in language education: exploring what it can offer

With artificial intelligence increasingly becoming an integral part of education, established ways of teaching and learning are being reimagined. To help understand this transformation, this section begins by outlining the ongoing technological shifts that impact teaching and by discussing their implications (Section 1.1.1). Following that, Section 1.1.2 takes an in-depth look into the potential of artificial intelligence to enhance language teaching and learning.

1.1.1. Education in the age of artificial intelligence

Although the use of digital technology to support language learning predates the emergence of AI (e.g., computer-assisted language learning, or CALL, developed as a

sub-field of language education as early as the 1980s; e.g., Guichon, 2024), the rapid spread of artificial intelligence and the speed with which it has permeated education have precipitated a visible qualitative change that necessitates rethinking how to best approach instruction, assessment and learner support.

While some technologies (for example, audio and audiovisual resources) were historically well established in certain settings, computer-based technology in language education (CALL) was often treated as a supplementary, optional strand rather than a baseline expectation for all teachers (Chapelle, 2001; Levy, 1997). Attempts to integrate digital technology in language education often aimed to generate and sustain learner motivation by infusing novelty and task variety into the learner experience. Such experimentation, though not rare, tended to be driven by individual initiative: in many contexts, innovations were predominantly introduced by relatively few teachers who had relevant training or interest, while others remained generally cautious and awaited evidence of effectiveness before adopting them. This dynamic produced a relatively measured pace of digital integration in language education, while implementation and outcomes varied considerably across settings (Goh & Aryadoust, 2025). Significantly, such usage was seldom initiated by learners, who were often reliant on teacher guidance not only for linguistic and pedagogical scaffolding but also for scaffolding in using relevant digital resources.

This situation stands in stark contrast with the current ubiquity of artificial intelligence. Recent studies (e.g., J. Freeman, 2025) have estimated that a majority of students actively engage with AI tools and frequently employ them for academic tasks. However, there is no convincing evidence that the use of AI aligns with sound pedagogical principles, and concerns have been raised that the prevalence of such *ad hoc* use, coupled with the absence of **critical** guidance, might deepen social, gender and professional inequalities (Révész et al., 2025). The cumulative effect of such concerns is that educators, including those who may have felt apprehensive about using technology in their practice, must increasingly reassess how they approach instruction, assessment and learner support.

Reassessing how instruction and learning are approached also means revisiting what counts as language use itself. Literacy, in an era of artificial intelligence, is likely to extend to the ability to interpret, mediate and generate meaning across diverse media (cf. Rowsell & Walsh, 2011), as opposed to narrower definitions of language learning, which confined themselves to developing communicative skills. This shift is usefully framed by the New London Group's (1996) account of multiliteracies, which placed design at the centre of literate activity, and which Kalantzis and Cope (2025) have since revisited in the light of generative AI. In this theoretical account, learners draw on "available designs" (i.e., existing conventions and discourses), transforming them into "redesigned" ones. AI output, a fluent aggregation of existing conventions, functions as an available design, which learners need to work through meaning-making practices such as prompting, interrogating and re-voicing. Such meaning-making involves engaging with language in diverse technologised forms, including skills for reading, interpreting, reflecting on, mediating and producing texts in multiple modalities (Chun et al., 2016). This constitutes a foundation for the development of pluriliteracies needed for deeper learning in different languages and cultures (Coyle & Meyer, 2021; Meyer et al., 2026).

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1.1.2. The case for artificial intelligence in language education

Seen against this backdrop of rapid technological change, evidence suggests that artificial intelligence can support language teaching and learning in important ways, provided it is thoughtfully deployed. For example, a recent systematic review of the efficacy and pedagogical implications of AI-driven chatbots in second language education (Wiboolyasarin et al., 2025) provides insights into how AI-assisted tools can support teachers in tasks such as providing personalised feedback, differentiated instruction and adaptive learning (Godwin-Jones, 2024). More broadly, artificial intelligence can enhance the language learning experience by enabling individualised learning paths, generating timely, individualised feedback and scaffolding in real time, extending access to language support beyond the classroom, and supporting repeated, low-stakes practice. Additionally, AI has been shown to foster increased learner engagement (e.g., Liu & Reinders, 2025; Yuan & Liu, 2025), a key predictor of success in language learning.

In more practical terms, AI tools can also serve as pre-writing brainstorming partners, virtual conversation simulators to develop speaking skills, including fluency, coherence, lexical and grammatical accuracy and phonological range, as well as to promote learners' willingness to communicate (Fathi et al., 2024; Hapsari & Wu, 2022). Additionally, they can function as reading companions, supporting learners in self-regulated, paced reading (Shafiee Rad, 2025). For learners with less developed proficiency in the target language, AI use has been associated with stronger writing self-efficacy (Warschauer et al., 2023; Zhang et al., 2025), and with support across multiple stages of academic writing, including outlining, revising, proofreading and structured reflection (Su et al., 2023).

Additionally, artificial intelligence applications can improve the quality of linguistic input to which learners are exposed. In some teaching contexts, educators may have limited access to up-to-date, varied and contextually rich linguistic resources for developing learning materials, including instructions, model texts and exercises. In such cases, AI tools can support materials development by expanding access to grammatically accurate, pragmatically appropriate and communicatively authentic language examples that complement teacher expertise. Similarly, in contexts where teachers lack confidence or expertise in generating communicative and task-based learning activities, artificial intelligence can empower teachers to move beyond overused (and often questionable) instructional formats, such as repetition, gap-fills or cloze activities, thus enabling learners to engage in more varied communicative and learner-centred alternatives (Wang & Tian, 2025).

In summary, AI is becoming increasingly relevant to language education, making it essential to develop a shared understanding of what it actually is and how it functions. The next section provides shared definitions and conceptual distinctions to support informed and consistent use of terminology in the discourse about language education.

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1.2. Understanding artificial intelligence in language education: exploring definitions, typologies and boundaries

To establish a shared conceptual background for the guidelines that follow, this section surveys the emerging field of AI for language education, by clarifying key definitions, outlining major categories of AI-powered applications, exploring how AI is being brought to bear in language teaching and learning and examining how the term itself has been used.

1.2.1. Towards a definition of artificial intelligence in language education

> I propose to consider the question, 'Can machines think?' This should begin with definitions of the meaning of the terms 'machine' and 'think'.
> Turing (1950, §1, p. 433).

Applications of artificial intelligence are rapidly becoming embedded in all aspects of life, from professional and scientific domains to personal interests and leisure activities. Even when we focus narrowly on education, or even more narrowly on language teaching and learning, as seen above, AI applications can provide support for a wide range of activities (Goh & Aryadoust, 2025), including conversational practice, text adaptation and translation, lesson planning, listening practice, communicative and form-focused tasks, reading comprehension, support with metalinguistic language explanations as well as assistance with assessment and feedback, and more.

One implication of the above is that, despite its widespread availability and apparent familiarity, AI resists a single settled definition. The instruments that currently shape policy and practice offer the following:

> AI systems are information-processing technologies that integrate models and algorithms that produce a capacity to learn and to perform cognitive tasks, leading to outcomes such as prediction and decision making in material and virtual environments [operating] with varying degrees of autonomy by means of knowledge modelling and representation and by exploiting data and calculating correlations. (UNESCO, 2021, para. 2(a))

> An AI system is a machine-based system that, for explicit or implicit objectives, infers from the input it receives how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments. Different AI systems vary in their levels of autonomy and adaptiveness after deployment. (OECD, 2024)

> 'AI system' means a machine-based system that is designed to operate with varying levels of autonomy and that may exhibit adaptiveness after deployment, and that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments. (European Parliament & Council of the European Union, 2024, Art. 3(1))

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> 'Artificial intelligence system' means a machine-based system that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations or decisions that may influence physical or virtual environments. (Council of Europe, 2024a, Art. 2)
>
> 'Artificial intelligence' (AI) encompasses a growing range of sociotechnical systems that: (i) comprise the interplay of people, institutions, policies, social practices and the environment, on the one hand, and computational devices, infrastructure, software, data and statistics, on the other, which together shape the design, development, deployment and use of such systems; and (ii) are designed to automate, augment or replace actions or tasks normally associated with human intelligence. (Council of Europe, 2026, para. 7.a)

The proliferation of definitional attempts is, in part, caused by the attempt to highlight different aspects of AI (as a system and as a field), depending on the needs and mission of various organisations. While such polyphony results in discourse about AI that is broader than it is precise, juxtaposing these definitions can reveal substantial common ground, which can be summarised in the following statements:

- AI systems are **machine-based** and operate by **inferring outputs** (predictions, content, recommendations or decisions) **from input** (Council of Europe, 2024a; European Parliament & Council of the European Union, 2024; OECD, 2024).
- AI systems have **various degrees of autonomy** (Council of Europe, 2024a; European Parliament & Council of the European Union, 2024; OECD, 2024; UNESCO, 2021).
- AI systems are designed to **automate, augment or replace tasks normally associated with human intelligence** (Council of Europe, 2026, para. 7.a.ii; UNESCO, 2021, para. 2(a)).
- AI systems are **sociotechnical** rather than purely technical, in the sense that they reflect the design choices and assumptions of their developers, and the dominant values of the contexts in which they are created (Council of Europe, 2026, paras. 7.a.i and 9).

Building on the observations above, and thinking specifically about the use of artificial intelligence in language teaching and learning, we define AI for language education as follows:

Artificial intelligence for language education refers to the use of AI-enabled systems and applications to mediate, support, augment or partially automate language teaching and learning processes, including teacher practices (for example, planning, materials design, feedback and assessment support) and learner practice (for example, rehearsal, interaction, drafting and revising, comprehension support and self-regulation), all with the aim of improving language learning conditions and outcomes, whether in teacher-guided instructional settings or self-directed learner contexts, and always in alignment with ethical governance principles.

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1.2.2. *Types of artificial intelligence*

In the previous section, artificial intelligence was viewed holistically, as a unitary entity. A distinction can be made, however, between three categories of artificial intelligence: generative, predictive and agentic AI. The differences between the two are summarised in Table 1 (below).

Table 1
Comparing types of artificial intelligence

	Generative AI	*Predictive AI*	*Agentic AI*
Function	Produces new content (text, images, etc.) which can support language teaching and learning	Analyses data to identify patterns and make forecasts and classifications	Pursues multi-step goals autonomously, taking actions across digital environments
Data use	Recombines and synthesises input to generate novel output (data patterns → new content)	Draws on historical data to predict outcomes	Uses real-time input and prior context to plan and execute sequences of actions
Typical tasks in education	Lesson planning support, materials generation and adaptation, dialogue simulation, writing scaffolding, feedback on learner output	Learner profiling, identification of strengths and growth opportunities, adaptive sequencing of learning content, support for placement / formative assessment	Autonomous tutoring sequences, adaptive curriculum management, multi-step task completion
Prevalence in language education	Increasingly used	Currently limited	Emerging; not yet widely adopted but rapidly developing

The first category, **generative AI**, constitutes a distinct subset within the broader field of artificial intelligence, characterised by its ability to employ generative models to produce novel content (OECD, 2026a). Some examples include ChatGPT, Gemini, Perplexity and others. This content emerges through the systematic identification and reproduction of

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patterns embedded in extensive corpora of training data, encompassing modalities such as text, images, audio and code. In the domain of language education, such output can take the form of syllabus frameworks, lesson plans, instructional materials, assessment rubrics and other pedagogical resources. Notably, generative AI has been mobilised with both productive and creative outcomes across educational contexts through applications including conversational agents (e.g., ChatGPT), image generation tools (e.g., DALL·E, Canva), and intelligent coding assistants (e.g., Claude Code).

Predictive AI, the second category, is a branch of artificial intelligence that identifies patterns in historical and usage data through statistical analysis and machine learning in order to predict future events or behaviours. Outside language education, predictive AI has seen applications in sectors like finance, healthcare and marketing, where it might be deployed to assess risks or predict customer behaviour. In education, predictive approaches have been less visible, in part because they often fall under regulated "high-risk" designations (European Parliament & Council of the European Union, 2024), although there is a growing interest in learning analytics and educational data mining, where predictive models are used to flag students at risk of non-completion, estimate likely performance, and recommend resources or next steps in their learning trajectory. One example of such systems is early-warning models, which draw on learner analytics to identify language learners who may be at risk of disengaging from a language programme, thus prompting intervention before difficulties become entrenched. A second example is recommender models, which use the same analytics to suggest readings, recordings or activities, and which may build pathways towards a target outcome on the basis of learners' prior performance and observed preferences.

A third category, **agentic AI**, has emerged more recently as a significant development in the field. Agentic AI refers to artificial intelligence systems designed to pursue specified goals through multi-step processes that may involve planning, decision-making, tool use, action execution, monitoring and adaptation over time. Unlike generative AI systems that primarily produce content in response to user instructions, agentic AI systems are oriented towards carrying out tasks in dynamic environments (for example, not only looking up the price of a train ticket but possibly using this information to book a train ticket for the user at the lowest price available on a specified website). Their autonomy can increase as they are given access to memory, tools, feedback mechanisms and delegated authority, though such autonomy must remain bounded by appropriate forms of human oversight, especially in educational contexts (OECD, 2026b). In language education, agentic AI remains at an early stage of adoption, but its potential implications are considerable: an agentic system might, for instance, autonomously design a sequence of learning activities, monitor learner progress, adjust difficulty and provide feedback, without step-by-step teacher input. This autonomy, while potentially powerful, also raises significant questions about oversight, accountability, and the appropriate boundaries of human delegation in educational settings.

It should be noted that the increasing autonomy of agentic systems is raising questions that the other two categories do not. When an artificial intelligence system is tasked with planning and executing a sequence of pedagogical decisions (e.g., designing a lesson plan or a handout) without step-by-step review, identifying where a human exercised judgment becomes more challenging, and it becomes correspondingly harder to establish who is answerable for ineffective or harmful practices or unfair outcomes. It is

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therefore imperative that teachers are meaningfully integrated in the loop (see Guideline 1 on legal and institutional responsibility, and Guideline 8 on teacher reflection and oversight).

. . .

The typology above serves to remind readers that artificial intelligence extends beyond familiar applications such as chatbots. However, the boundaries between these categories are porous, and individual applications frequently combine them, with agentic capabilities increasingly being bundled into general-purpose assistants.

1.2.3. Functions of artificial intelligence in language education

A different way to categorise artificial intelligence applications would be with reference to the range of affordances that these create for teaching and learning. This range can be conceptualised as a continuum, ranging from "narrow" to "broader". However, for descriptive purposes, the discussion that follows (and Table 2, below) will reduce this range into a set of provisional categories, which should be understood as overlapping and porous rather than mutually exclusive. It is also important to remember that many applications evolve by adding features that expand their scope, or – less commonly – by creating specialised services that take over part of their functions.

Table 2
Artificial intelligence applications based on range of affordances

	Core function	*Examples of current applications*
Artificial narrow intelligence	Limited support for specific tasks	Antidote, Grammarly, Elsa, Quizlet
Intelligent tutoring systems	Adaptive instruction & feedback	Duolingo, Knewton, Write & Improve
Conversational agents	Simulated real-time interaction	ChatGPT, Gemini, TalkPal
Generative platforms	Creative content production	ChatGPT, Gemini Notebook
Assessment & analytics	Learner performance tracking	Microsoft Reading Progress, WriteToLearn

Artificial narrow intelligence (ANI) or "weak" AI is a broad term that includes specialised applications, such as grammar and style checkers, vocabulary learning applications, or applications for improving pronunciation. Such applications can help with providing micro-skill support and immediate feedback.

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Intelligent tutoring systems (ITS) are used to deliver scaffolded language learning content to learners, using personal learning pathways and adaptive feedback. This allows learners to learn in a self-paced manner that is tailored to their strengths, needs and interests.

Conversational agents (chatbots) simulate real-life dialogue by generating conversational prompts and appropriate responses. Using such applications enables learners to practise speaking, writing and interaction in a multitude of languages where finding conversational partners would be challenging, and it also provides a "safe space" for practice.

Generative platforms are tools that leverage generative AI capabilities (see above) to produce original content for language learning purposes. Some examples of such usage include creating texts with which learners can interact, adapting existing texts for pedagogical purposes, and reformatting content (e.g., from video to text). Generative AI can be used, among other things, to support open-ended tasks and differentiated instruction.

AI-assisted assessment and analytics applications can make inferences about learners' profiles including identifying both strengths and potential areas of difficulty. By leveraging advanced data analytics, these applications may provide diagnostic reports or formative assessments to inform and support language learning.

As above, it should be stated that the boundaries in this typology are shifting as well. This is happening because applications that began as single-purpose tools are progressively integrating general-purpose generative capabilities. For instance, AI-assisted grammar checkers (e.g., Grammarly and its French-language counterpart Antidote) currently offer open-ended drafting and rewriting in addition to their original proofreading functions. It should be obvious that teachers and learners should therefore evaluate what a tool currently does rather than rely on how it was first classified.

1.3. Using artificial intelligence: surveying existing guidance

The previous section concluded with the observation that the rapid development of artificial intelligence creates a need for domain-specific guidance to support its responsible and pedagogically sound use. It is therefore unsurprising that several institutions and organisations worldwide have taken up the call and promulgated guidelines to address the multifaceted challenges posed by AI technologies, alongside several initiatives designed to help teachers develop AI literacy and navigate their evolving roles (Kohnke et al., 2025; Long & Magerko, 2020). While a comprehensive survey of all such guidelines, even if possible, would fall outside the scope of this document, this section will present some of the key principles and concerns shaping the discourse about the use of AI in language education.

This discussion is structured into three stages: relevant guidelines are surveyed first, leading to the presentation of key themes, and finally, a problematisation of questions and challenges that remain to be addressed.

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1.3.1. Survey of guidance on the use of artificial intelligence in education

This section presents an overview of legislation, institutional guidelines and other advice on the use of AI in educational contexts. To help contextualise the guidelines that are presented in the second part of this document, the discussion draws on guidance spanning a broad range of educational contexts rather than focusing exclusively on language teaching and learning. A searchable index of these documents can be found in the AI Lang Moodle course.

The international landscape Several attempts to regulate the usage of artificial intelligence have originated from intergovernmental or supranational entities. For example, the United Nations Educational, Scientific and Cultural Organization's *Recommendation on the Ethics of Artificial Intelligence* (UNESCO, 2021) presents a comprehensive human rights-based approach. Key tenets include proportionality, safety and security, privacy and data protection, human oversight, transparency, accountability, sustainability and the promotion of multi-stakeholder governance. Similarly, the Organisation for Economic Co-operation and Development (OECD)'s *AI Principles* focus on fostering innovative and trustworthy AI while upholding respect for human rights and democratic values. Additionally, the Council of Europe *Framework Convention on Artificial Intelligence and Human Rights, Democracy and the Rule of Law*, adopted in 2024, represents a milestone in global AI regulation. Some key provisions are that AI systems must respect human rights, the rule of law and democratic values throughout their life cycle; that all parties should assess and, where relevant, mitigate risks caused by the use of AI; that there should be human oversight, transparency and accountability in the use of AI systems; and that AI use promotes safety, non-discrimination and privacy. Such regulatory attempts, which vary in their prescriptive force, supplement existing legislation (e.g., the *General Data Protection Regulation* in the European Union, or the *Personal Information Protection and Electronic Documents Act* in Canada), which predates the widespread use of artificial intelligence, but remains relevant.

The Council of Europe Education Strategy In parallel with the above, the Council of Europe is actively engaging with the emerging challenges and dilemmas associated with the educational deployment of AI technology, as part of its Education Strategy 2024–2030 (Council of Europe, 2024b). While artificial intelligence is not explicitly mentioned as a focus point, the strategy provides a broad frame for its integration into education. This frame highlights dimensions such as the democratic mission of education, social responsibility and a human rights-based approach to digital transformation. The strategy also calls for new legal and policy frames, reference standards and capacity-building tools for the new educational realities that are emerging.

This strategy builds on a strong empirical foundation, namely an international survey on the state of artificial intelligence in education (Chounta et al., 2024). The results of the survey indicate that although general policies and strategic guidance for AI exist in many European countries, specific policies for AI and education have yet to be developed, although in more recent years there has been an acceleration of policy frameworks addressing the role of artificial intelligence in education. Despite that, what remains conspicuously absent are the accompanying resources and, crucially, the practical

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demonstrations and teacher training modules that would illuminate how these policies can be meaningfully implemented and consistently upheld in real classroom contexts.

The strategic vision of the Council of Europe is further articulated in the Artificial Intelligence and Education project. One of the main outputs has been a preparatory study for the development of a legal instrument that regulates the use of AI in education (Havinga et al., 2024). Key elements of this policy include (a) the priority of human rights, (b) transparency and oversight, (c) accountability, (d) democratic principles, (e) inclusivity and equitable education, and (f) shared governance mechanisms. This is intended to ensure that the use of artificial intelligence is done within the frame of a principled, robust and flexible regulatory response.

This preparatory work has since culminated in a formal policy instrument, the **Council of Europe Recommendation on AI literacy** (Council of Europe, 2026). The Recommendation defines AI literacy as "the capacity to engage with AI critically, ethically, responsibly, safely and effectively" along three dimensions: the human, the technological and the practical, with particular emphasis on the first. The Recommendation is articulated in the form of guiding principles, several of which bear directly on language education. Some of the most important are: that AI is sociotechnical rather than neutral; that AI systems may threaten diversity by reinforcing cultural and linguistic biases and standardisation; that human agency, judgment or responsibility must be preserved rather than delegated to automated systems; and that individuals should be enabled to make informed decisions about their use *or non-use* of AI. Because it is addressed to member states rather than to individual language teachers, the Recommendation stops short of specifying what AI literacy might look like in a language classroom. The framework presented in the second chapter of this document can be read as one domain-specific response to that gap.

Local initiatives The exponential growth of artificial intelligence has also seen the generation of several documents (white papers, position papers, etc.) that seek to regulate the use of AI when working with humans (e.g., Foltýnek et al., 2023; Green et al., 2025; Hollanek & Sobey, 2025). One common theme in such guidance is the emphasis on human-centred and value-led design, underscoring themes such as dignity, agency, autonomy and well-being, as opposed to a singular focus on technical efficiency. The risks of manipulation and emotional dependency are also noted, alongside the technical limitations of present-day AI applications (e.g., hallucinations, lack of contextual relevance and understanding, bias, etc.). Building on this, the consensus seems to be that artificial intelligence can be best employed as a supportive or augmentative tool within the frame of clear ethical and regulatory standards.

More specialised guidance, in the field of higher education, additionally addresses concerns centred on academic integrity and responsible use. The guidelines published by Harvard University, for instance, emphasise issues such as data protection, content accuracy, intellectual property rights, phishing awareness and the recognition that public AI tools are not inherently private. Similarly, guidance by Stanford University and the Massachusetts Institute of Technology (MIT) underscore the importance of academic integrity, advising against the undisclosed use of AI for assignments and exams, providing guidance for instructors, and prioritising ethical AI use and data protection. ETH Zurich

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guidance also aligns with these principles, advocating responsible, fair and transparent AI utilisation, mandatory disclosure of AI tools, and general adherence to established ethical guidelines.

Emerging policies also take into account the learning affordances provided by artificial intelligence. For instance, the Russell Group, a collective of leading UK universities, has developed a set of principles for generative AI in education, stressing the need for AI literacy among students and staff, adaptable teaching and assessment methods, maintenance of academic rigour and integrity and the fostering of collaborative best practices. Similarly, other university systems have been developing policies for the effective use of artificial intelligence, such as the ETHICAL Principles AI Framework for Higher Education (Wynants et al., 2025). In a similar vein, bottom-up initiatives, such as the recommendations by the Lancaster University Students' Union highlight the potential for ethical integration of AI into student assessments.

In summary, while the specific guidelines vary in their granular focus, a common thread among diverse organisations in Europe and North America is the imperative for AI to be developed and used in a manner that is human-centric, transparent, accountable and respectful of fundamental rights and societal values. This common ground is explored further in the next section.

The AI Lang course contains a searchable index of the relevant policy instruments, institutional guidelines and position papers, including the ones surveyed here. This includes brief descriptors indicating the scope, issuing body and jurisdiction of each document, as well as links to the original document.

1.3.2. Shared concerns

In the existing guidance for AI in education (surveyed in Section 1.3.1), certain themes appear to recur, even if their relevance across contexts varies. One of the most salient concerns is the factual **accuracy** of AI output. The large language models (LLMs), on which many contemporary AI tools rely, are fundamentally probabilistic systems (or "guessing machines"). This means that while they can generally produce output that is highly plausible, based on a series of complex statistical and algorithmic calculations in the context of a conversation, they may not always produce the most accurate or reliable results, particularly when the optimal response is statistically less probable. In some language learning activities, such as those focused on linguistic form, conversational fluency or creative writing, absolute factual accuracy on non-linguistic facts might be of secondary importance. By contrast, in contexts where content is part of the learning objectives, as is the case, for example, with Content-Based Instruction (CBI) or Content and Language Integrated Learning (CLIL) approaches, inaccurate or sometimes entirely fictitious output ("**hallucinations**") can be a serious concern. While it is likely that future AI applications will reduce the frequency of hallucinated content, existing guidance emphasises that educators and students should remain epistemically vigilant, treating AI-generated content as provisional and subject to verification. In practice, this means interrogating and validating claims against reputable external sources, especially when users lack the disciplinary expertise needed to evaluate accuracy independently.

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A second area of concern in the existing guidance is **privacy**. Many AI applications, including educational ones, operate by collecting data (e.g., voice recordings, writing samples, records of chat conversations) and **metadata** from teachers and learners. Such user-generated input might be retained for the purpose of generating personalised learning experiences, but can also be used by the companies to help train and refine their large language models. Consequently, educational guidelines must clearly inform educators and students about the data they share with AI platforms, how this data is maintained and managed, and the risks associated with its potential misuse. They must also address the dangers of unauthorised disclosure, whether deliberate or inadvertent, by the companies hosting these systems. Such risks are particularly grave when learners are minors or members of vulnerable populations, as is often the case in language education. Similarly, schools, administrative authorities and other decision-making entities need to exercise due diligence when selecting tools that are to be integrated into schools, favour resources that protect to the maximum level possible the privacy of the users (teachers and learners) who will be using them, and communicate information transparently to the end users and other stakeholders.

Another concern that features prominently in guidance for the use of AI is **fairness and representation**. As noted above, AI systems depend on large datasets, from which they draw text or visual elements in order to produce responses. Such datasets are likely to reflect existing societal biases, and a common concern therefore is that the use of AI is likely to reproduce, and possibly amplify, prejudicial content. This could result in the production of learning materials that contain unrealistic representations of the social world or the privileging of dominant language varieties (Bender et al., 2021).

As can be inferred from the discussion above, many of the concerns that are voiced in the existing guidance on AI in education are applicable to language teaching and learning. However, in many cases, the ways in which such concerns might be manifested are domain-specific. For example, concerns over privacy are obviously relevant across educational contexts (and outside education), but language education often encourages sharing of personal narratives or voice samples. Similarly, the risks associated with representation might manifest in language education through practices that perpetuate linguistic hierarchies, marginalise less widely spoken languages and varieties, and undermine efforts to promote plurilingualism.

Furthermore, the activities often used in language teaching and learning can also introduce domain-specific challenges. For instance, when students or teachers use AI applications to scaffold speaking or writing tasks (e.g., dialogue creation), adequate pedagogical framing is necessary to encourage creativity and avoid the learning of formulaic language. Similarly, when it is used for the purposes of a placement or aptitude testing, predictive AI might produce skewed assessments of linguistic proficiency. A persistent problem in language assessment is that it tends to privilege aspects of communication that are readily quantifiable (e.g., grammatical accuracy), at the expense of those that are more challenging to effectively operationalise, such as pragmatic competence, intercultural awareness, or creative language use (Bachman & Palmer, 2010). What is especially concerning, in the case of AI-assisted assessment is that such algorithmically generated evaluations might be framed in an aura of authority that

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discourages critical scrutiny and appeals, especially when assessing productive skills (Warschauer & Grimes, 2008).

In sum, guidance on the use of AI in education consistently emphasises the importance of approaching tools with a high degree of **critical** awareness and educational sensitivity. Such concerns are intensified in language teaching, where linguistic diversity, cultural nuance and learner identity are central themes. A number of unresolved challenges will be explored in the next section (Section 1.3.3), where the focus shifts from identifying risks to examining how AI can be harnessed to support language learning without compromising ethical, pedagogical, and social values.

1.3.3. Open questions and challenges

While the commonalities listed above suggest a broad consensus regarding at least some aspects of AI, such as transparency, fairness, privacy and safety, there is still considerable ground where perspectives diverge and practical implementation remains uncertain.

For instance, while awareness of risks associated with the unregulated use of AI is growing, balancing efficiency gains with **privacy rights** complicates AI deployment – or, to frame this in a way more relevant to language education, there are still unresolved issues concerning who might benefit from the efficiency gains associated with AI usage, how such added efficiencies might threaten job security, and how the pedagogical benefits of AI-supported language learning balance with rights to privacy.

Furthermore, the question of how the responsible implementation of artificial intelligence can be used not only to prevent harm and mitigate risks but also to contribute to **social good** remains open. In the context of language education, some potential uses (such as providing access to practice opportunities, which may not be available otherwise) offer considerable promise, and the challenge that remains is to ensure that such affordances are utilised in ways that are effective and ethical.

While AI excels at delivering personalised content and feedback, its inherent design can inadvertently circumscribe **learner autonomy** by channelling users into pre-structured pathways, thus limiting genuine exploration and the spontaneous, creative use of language, especially when AI-driven systems prioritise efficiency and assessment over openness and flexibility (Brod et al., 2023). This means that AI systems that pre-structure learning pathways in a very restrictive sense may shift learner motivation from authentic language use to language use rewarded by the system, with long-term implications for sustained engagement. Of course, this limitation is not absolute. The extent to which AI constrains or fosters autonomy depends on its underlying design, pedagogical philosophy and intended use. It also relates to the users' competence in writing effective prompts that align with intended purpose and ethical use. Ongoing research in AI for education increasingly focuses on balancing personalisation with autonomy, agency and opportunities for creative language use.

Furthermore, AI may be able to provide useful **feedback** in many cases, which might compare favourably to the kinds of feedback provided by less experienced teachers or

teachers working under time constraints, but its agency is purely algorithmic (Kherbouche, 2026). This means that it lacks, by definition, the nuanced understanding of a skilled and experienced human educator to adapt to unexpected learner needs or encourage **critical** reflection beyond programmed responses. This can lead to a deskilling of learners, as their reliance on AI for prescriptive solutions might diminish their capacity for independent problem-solving and critical linguistic analysis. Consequently, the development of independent critical thinking and self-directed learning in complex, ambiguous linguistic situations can be inadvertently stifled, especially through the use of generative AI (Alm, 2024).

Another concern, which is perhaps less visible in existing policy guidance but increasingly salient in research and in teachers' own accounts, relates to **cognitive decline**. In situations where AI supplies answers rather than pedagogical scaffolding, learners could disengage from the effortful work on which language learning depends. Such a scenario would involve offloading the retrieval, formulation and revision that produce learning gains, substituting AI use for the productive struggle that language development requires. This risk makes task design, rather than prohibition of the tool, the locus of the pedagogically appropriate response (see Guideline 7).

Finally, it is important to consider the **added pedagogical value** of artificial intelligence, i.e., what AI adds relative to non-AI alternatives. Many of the affordances offered by AI – such as instant feedback, adaptive exercises or multilingual glossaries – have partial equivalents in traditional teaching, textbooks, or peer interaction. AI becomes pedagogically meaningful when it enhances these affordances, extends learner autonomy, or enables activities that would otherwise be impossible or prohibitively resource-intensive. Comparing AI-supported approaches with existing methods helps educators identify when AI use is justified and when simpler, low-tech solutions may suffice.

Other issues that are emerging in public discourse about artificial intelligence include the lack of **transparency** inherent in the 'black box' model of AI (Memarian & Doleck, 2023; Jiang & Zhang, 2025), the question of **bias** and the **negative environmental impact** (De Vries, 2023; Luccioni et al., 2023).

Such persistent challenges suggest that the integration of AI in language education requires careful consideration. Some lines of thought include the need to ensure fairness, transparency and accountability, as well as alignment with inclusive, socially equal educational policies. Used with care, AI has the potential to support, rather than undermine, intercultural teaching and mutual respect across cultures, and to assist in establishing respect for difference as an integral part of classroom dynamics.

The definitions and concerns outlined in this chapter provide a foundation for understanding the role of AI in language education. The second part of this document (Chapter 2: Framework for the ethical use of AI in language education) will build on this conceptual groundwork, and make recommendations for how AI can be implemented in

ways that align with core pedagogical values and priorities. These recommendations are intended as flexible reference points, and should be interpreted in connection with the points raised in this section, as well as the particularities of teachers' contexts.

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Part 2

Framework for the ethical use of AI in language education

2. Framework for the ethical use of AI for language education

2.1. Scope and structure of this framework

This chapter presents a framework of principles, guidelines, competence descriptors and reflection tasks that are intended to support the use of artificial intelligence in language education.

More to explore

https://moodle.ai-lang.ecml.at/

This guidance is most directly addressed to teachers of second and foreign languages, as well as teachers involved in heritage language education, who can use the framework to problematise their practice and focus their professional development. To assist with the latter, the framework interconnects with the **AI Lang course**. Secondary audiences include: (a) teacher educators, who can use the framework to extend and enrich language teacher professional development, and to inform assessment in language teacher education; and (b) policy planners and administrators, who may find the framework useful in developing accountability structures for the use of AI-assisted language education. The ultimate beneficiaries are language learners. This includes learners in self-directed language learning situations, who may benefit from selected elements of the guidance to make informed and **critical** use of AI tools. It also includes learners who are vulnerable, or may be rendered vulnerable through the uncritical adoption of AI-assisted practices. For these learners, the framework provides a safeguard and a resource: a safeguard in the sense that the framework explicitly foregrounds their rights, identities and well-being as non-negotiable considerations in any pedagogical decision about AI use; and a resource in the sense that, where appropriate, it can support their own informed and critical engagement with AI tools.

2.1.1. How this framework is organised

The guidance that is presented here for the use of artificial intelligence in language education is organised hierarchically, as shown in Figure 1. At its core lie four **principles** derived from the emerging consensus on how artificial intelligence should be used (see Section 1.3.2 above). These principles (e.g., "The use of AI in language education should be responsible") are discussed in more detail in the following section. The function of these principles is to situate the framework within a broader commitment to humanistic and critically informed language education, and to anchor the guidance to the core values of democracy, human rights, and multilingualism. The principles are the highest level of abstraction in the framework, and are expected to remain relevant even in the face of technological change.

Each principle is operationalised into **guidelines**, which are presented separately later in this document (Sections 2.3 et seq.). The eight guidelines that make up the framework translate the broad value orientations of the corresponding principle into specific domains of professional practice (e.g., "Safeguard the data security and digital safety of all participants"). In other words, the guidelines give the principles practical shape, by identifying the areas of competence and responsibility that language teachers need to attend to when using AI. Like the principles, the guidelines are intended to be broadly

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applicable across different educational contexts and levels, and to retain their relevance as the technological landscape continues to evolve.

Figure 1

Structure of the framework

...

A set of several **competence descriptors** is derived from each guideline. These descriptors define what ethical AI use looks like in observable, practitioner-level terms. Building on the definition of competences by Vitello et al. (2021, p. 4), these competences refer to language teachers' ability to integrate and apply knowledge, skills and psychosocial factors (e.g., beliefs, attitudes, values and motivations) that are relevant to their teaching context in order to consistently perform successfully within AI-assisted language education. These descriptors have been reformulated in question format (e.g., "*To what extent do the AI-assisted tasks I design increase learner agency?*"), to avoid the binary framing implicit in 'I can...' statements. The function of the competence descriptors is to make the guidelines actionable, by offering language teachers a concrete basis for self-assessment and professional development. Unlike the principles and guidelines, the competence descriptors are more context-sensitive, and teachers are encouraged to prioritise those that are most directly relevant to their role, their learners, and their institutional setting.

Finally, each descriptor is further illustrated by a number of **reflection questions**, which invite teachers to connect the framework to the particularities of their own practice (e.g., "Does the AI tool you use require learners to sign in? If so, where can you find out what data it collects?"). The function of the reflection questions is not to elicit predetermined correct answers, but to scaffold the kind of professional judgment, critical awareness and contextual sensitivity that responsible AI use requires. They can be used for individual reflection, for structured professional dialogue with colleagues, or as a basis for participatory discussion with learners and other stakeholders, bearing in mind that the ecosystem of AI applications and regulatory guidance is likely to evolve dynamically.

2.1.2. How this framework can be used

The guidance in this framework is intended to provide language teachers and administrators with a set of reflective tools, which they can use to develop contextually appropriate practice, depending on the affordances and constraints of their settings. It does not constitute a compliance checklist or an inventory of conditions that must be fulfilled in order to integrate artificial intelligence into language education.

The reflection questions that are included under each guideline are central to this perspective. Teachers are encouraged to use and adapt these as necessary to help scaffold structured individual reflection, collaborative professional dialogue, and, where appropriate, participatory discussions with learners, school leaders, data protection officers, IT personnel and other stakeholders. These questions are not meant to elicit predetermined ("correct") answers; rather, they serve to support informed professional judgment by surfacing assumptions, clarifying values, and making explicit the pedagogical, ethical, and institutional considerations that shape decisions about AI use.

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It should also be noted that, in many cases, the burden of ensuring ethical use, as articulated in this document, is shared. For instance, considerations of how data are used by artificial intelligence tools often presuppose a degree of legal and technical sophistication that is unrealistic for language teachers. The guidelines should therefore not be interpreted as an attempt to individualise responsibility, but rather as tools for drawing attention to where gaps in expertise, policy, or institutional infrastructure may exist, and to direct collective attention to addressing such gaps and developing informed, joint decision-making.

In this sense, the guidelines are expected to serve as a reflective instrument and as a diagnostic tool: they help practitioners recognise the limits of individual agency and provide a structured basis for collaborative deliberation on the ethical use of artificial intelligence in language education.

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2.2. Principles underpinning the ethical use of AI in language education

The guidance that is presented below is premised on four foundational principles that reflect the emerging concerns about the affordances and potential pitfalls associated with the use of AI. These are that the use of artificial intelligence in language education should be (a) **safe**, (b) **responsible**, (c) **purposeful**, and (d) **human-centred**. These are outlined below:

I. AI-assisted language education should be **safe**. All participants in education (learners, teachers and institutions) must be protected from actual or potential harm, both immediate and long-term, when engaging in AI-supported language learning. This is done by ensuring compliance with applicable national and supranational regulation as well as institutional duty of care. This principle can be operationalised in the following guidelines:
   1. *Comply with existing legal requirements and institutional standards*.
   2. *Safeguard the data security and digital safety of all participants*.

II. AI-assisted language education should be **responsible**. This principle positions artificial intelligence in language education as a force for social good, by fostering linguistic and cultural diversity, and by minimising environmental harm and social injustice. In practice, this principle can be unpacked in the following guidelines:
   3. *Promote social justice, linguistic and cultural diversity*.
   4. *Foster environmental sustainability*.

III. AI-assisted language education should be **purposeful**. Focusing on the pedagogical integration of artificial intelligence, this principle entails that AI should produce substantive added value to the language learning experience and enhance sound pedagogical practice. In other words, it should create affordances for teaching and learning that would not be possible, or would be prohibitively demanding in resources, without the use of such technology. The guidelines that operationalise this principle are the following:
   5. *Enhance theory- and policy-informed language pedagogy*.
   6. *Create expanded language learning opportunities*.

IV. AI-assisted language education should be **human-centred**. What this principle indexes is that both language learners and language teachers can draw on existing traditions in language education, together with the affordances of artificial intelligence, in order to become more empowered agents. The guidelines through which this principle is operationalised are:
   7. *Prepare learners for critical, healthy engagement with AI*.
   8. *Empower language teachers as reflective professionals*.

The use of AI as described in the framework should be understood as a holistic ethical practice. The principles and guidelines that make up the framework are presented separately for reasons of analytical convenience, but they overlap conceptually (e.g., considerations of social justice cannot be disentangled from guidance on safety). The guidelines therefore should be read as showing differences in nuance and emphasis rather than categorical distinctions.

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[Insert diagram here]

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2.3. Principle I: AI-assisted language education should be safe

Are our uses of artificial intelligence in language education legally compliant, transparent and low-risk?

Establishing legal compliance and conformity with institutional duty of care are the minimum conditions for the safe integration of artificial intelligence in language education. This involves compliance with applicable regulatory frameworks, including local, national and transnational legislation (in the EU context, this includes especially the General Data Protection Regulation and the EU Artificial Intelligence Act) and any other legislative provisions regarding data privacy and security. Additionally, this principle covers any other measures necessary for protecting the physical, psychological and digital well-being of participants, including especially underage learners and other vulnerable individuals.

To ensure safe use of AI, language teachers and learners should be mindful of (a) how the data they intentionally provide (e.g., while registering to use an application) as well as data that are incidentally collected (e.g., through cookies) are used; (b) how input that they provide during educational activities is used; and (c) what the applications' data retention policies are.

All these data collection and retention policies must comply with relevant legislation (Guideline 1), a requirement that is particularly important as language education often involves minors and members of vulnerable populations. However, since technological advances are likely to outpace legislative work, teachers and schools should also exercise care that extends beyond minimum legal requirements to proactively safeguard the data security and privacy of users (Guideline 2).

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Guideline 1: Comply with existing legal requirements and institutional standards

Although competent authorities, both at national and supranational (e.g., European Union) level, are beginning to address some challenges associated with the growth of artificial intelligence (see Section 1.3), the use of AI in education remains largely untested legal territory (Havinga et al., 2024; Holmes & Porayska-Pomsta, 2023). As language teachers cannot yet confidently rely on conventional practice or established legal precedent, caution and informed judgment are necessary to ensure that their practices are compliant with any existing legislation and consistent with the policies and standards of their institution. Some considerations they need to bear in mind relate to terms of service and age restrictions, copyright and intellectual property, and accountability and transparency. Sometimes, institutional policies might provide guidance on some or all of these areas; where such policies exist, teachers should familiarise themselves with them and ensure that their use of AI tools aligns with institutional expectations.

Regarding possible **restrictions**, teachers should ensure that the use of AI tools in language education does not result in students being exposed to content or interactions that are prohibited by local law. Additionally, they should be aware of any restrictions in the legally binding terms of service of the applications they use, such as terms that might require paid plans for institutional or professional use or terms that restrict use to older individuals (typically 13+ or 18+). Institutional procurement or IT policies may also restrict which tools are approved for use with learners. Teachers should also ensure that learners are legally competent to consent to the terms of service (or, alternatively, they should seek permission from their parents or guardians), and that users or their guardians are proficient in the language in which the terms and conditions are stated, in order to ensure informed consent.

The ability of generative AI to generate, remix or translate content such as text, images and videos can also give rise to legal concerns, especially when AI tools are used to produce simplified or translated versions of copyrighted content. While legal provisions regarding **copyright** vary, sometimes permitting limited educational use, language teachers should nevertheless ensure that all AI-generated content used in class is properly attributed and that ownership of intellectual property (including that of the learners) is respected. Where policies on academic integrity or intellectual property exist, which address and identify AI-generated content, these should inform teachers' practice. This can be done, for example, by adding information about the original source and the methods used to adapt the text. Schools and education systems might need to generate policies to ensure consistent application of these practices.

A third set of legal concerns relates to accountability and transparency, or what is sometimes termed the **Fairness, Accountability, Transparency and Ethics** (FATE) requirement (Memarian & Doleck, 2023). Whether planning learning experiences or engaging in assessment, teachers should be aware of the complexities associated with delegating tasks to non-accountable agents, such as agentic AI systems that can complete tasks or make decisions without being held legally, ethically or professionally responsible for the outcomes. In other words, teachers must be aware of their own responsibility for poor learning outcomes, unfair assessments, or inappropriate learning experiences resulting from automated methods. Institutional accountability frameworks, where they exist, should make explicit how responsibility is allocated when AI is involved in consequential decisions about learners.

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Competences	Reflection questions
1. [**I can understand and use AI tools lawfully**] *How fully do the AI tools I use align with local, national and institutional legal frameworks?*	- *Where can I verify that a tool meets legal or institutional standards?* - *Do I check and summarise the main points of a tool's terms before students use it?* - *How do I ensure learners (or guardians) understand what they agree to?* - *What steps do I take to prevent exposure to illegal or harmful content?*
2. **[I can protect intellectual property]** *How carefully does the content I (co)generate with AI adhere to intellectual property laws?*	- *How do I attribute adapted or AI-generated materials? Would my attribution practices hold up if questioned by a colleague, a learner, or a rights-holder?* - *How do I safeguard learners' creative work from being reused without consent?* - *How am I modelling the intellectual property practices I expect from my learners when they use AI?*
3. **[I can maintain responsibility for the use of AI in my teaching]** How do I maintain responsibility as a teacher when delegating tasks to AI applications?	- *Why do I need AI to assist in this task, instead of relying on other tools?* - *When AI generates feedback, or learning materials, what does my review process look like before that output reaches learners?* - *How would I explain my use of AI to learners or colleagues?* - *How do I provide space for learners to challenge AI-generated output?* - *How am I modelling the transparent use of AI in my teaching?*

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Did you know?

- The 2024 EU Artificial Intelligence Act prohibits the use of AI systems for emotion recognition and real-time facial recognition for monitoring/discipline (Article 5). Tools that manipulate student behaviour or exploit vulnerabilities are similarly not allowed. Any system that creates or expands facial recognition databases from internet images or CCTV is also expressly prohibited.
- High-stakes activities, including automated grading, admissions and other activities that can have a significant impact on student opportunities are also regulated under Annex III (which classifies such procedures as high-risk) and Articles 9-16. Such activities require strong human oversight and explainability, risk mitigation protocols (Article 9), technical documentation and logging of system behaviour (Articles 11-12), transparency towards users (Article 13), and meaningful human oversight (Article 14). Providers and deployers of high-risk systems are also required to implement clear appeals processes for students and to maintain accountability documentation under Article 16.
- Additionally, whenever learners interact with any AI system such as a conversational agent or a generative AI application, or engage in similar "limited-risk" activities, educators are required to clearly disclose that the agent is not human, unless this is clear from the context (Article 50).

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Guideline 2: Safeguard the data security and digital safety of all participants

Education providers have an ethical responsibility, or duty of care, to ensure the privacy and data security of people involved in language teaching and learning. Existing legislation, some of which predates the widespread adoption of artificial intelligence, generally provides robust frameworks, but cannot fully anticipate the pace and complexity of technological development. This means that legal compliance, though necessary, does not always provide sufficient protection for users in language education, where activities routinely involve the sharing of personal narratives, voice recordings and written samples, and where learners are often minors or members of vulnerable populations whose data warrants heightened protection.

Artificial intelligence applications function by **collecting learner data**, such as voice recordings or writing samples, and reacting to such input. For example, when a learner engages with a conversational agent for speaking practice, conversations about personal experiences may be retained; similarly, texts on controversial topics, produced for developing writing skills, might be retained by AI-supported feedback applications. Applications might also harvest data beyond what is strictly required for educational purposes, such as personal identifiers (cookies), device information, or engagement and behavioural metrics (Liu & Khalil, 2023). Such practices are generally legal, assuming informed consent (see Guideline 1). However, digital tools need to be vetted to ensure that their data collection practices align with pedagogical needs and that issues of informed consent are appropriately addressed and documented.

Educators should also be aware that user content (e.g., prompts, voice samples) and **metadata** might be retained in ways that are hard (or, at times, impossible) to erase. This can be problematic, as obsolete or poorly managed data storage systems may pose privacy risks. Even anonymised data might be compromised through accidental exposure or used for profiling (e.g., revealing association with minoritised linguistic groups, learning differences, or non-academic attributes like sexual, religious and political affiliations). Especially when working with vulnerable individuals, such as underage learners, migrants, refugees and asylum seekers, AI use is best limited to trusted applications, and users need to be educated about data security best practices, such as minimising the disclosure of personally identifiable information in prompts and uploads, avoiding the sharing of identifiable work and reporting suspected breaches promptly through established institutional channels.

Another issue that warrants careful consideration is the possibility of **data mining**, an issue not restricted to AI applications, but amplified in them. Such practices become problematic when data generated in educational settings are repurposed for unauthorised machine training, commercial or even surveillance purposes. Data mining can also be an issue when an application uses user-provided data (and possibly metadata) for training the AI model or learner **profiling**. While such profiles can help personalise learning, they also raise questions of consent and representation. They can also exacerbate linguistic bias, and (where AI-assisted assessment is allowed) they might result in flawed assessment outcomes. Educators need to monitor how learners are guided through content, and they should use their professional judgment to adapt or override AI suggestions.

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Competences	Reflection questions
4. **[I can identify what types of data are collected by AI applications, which might be misused]** *How confidently can I identify what types of data (e.g., voice recordings, writing samples, **metadata**) are collected when using AI applications in language education?*	- *What routines do I have (or can I put in place) for checking whether AI tools require learners to sign in, and (if so) what data they collect?* - *How do I decide whether data collection by AI applications is pedagogically needed and proportionate for language learning purposes?* - *What procedures do I have (or can I put in place) for deleting user data or preventing it from being used for purposes beyond the classroom?* - *How might I adapt the use of AI tools to protect learners' rights and identities?*
5. **[I can raise awareness about the data AI applications gather and its possible uses and misuses]** *How successfully can I raise awareness of learners, colleagues and parents about risks to data privacy?*	- *Are there learners in my classroom whose personal data could put them at risk if mishandled?* - *How might I explain data privacy risks to learners or colleagues in accessible and practical terms?* - *How might I involve learners in reviewing the privacy settings of a tool we use together?* - *What should I discuss with my class or team about the implications of learner profiling?*

Did you know?

Some relevant legal documents that regulate data privacy include the following:

- The European Union AI Act (European Parliament & Council of the European Union, 2024) [EU]
- The General Data Protection Regulation, or GDPR (European Parliament & Council of the European Union, 2016) [EU]
- The Family Educational Rights and Privacy Act, or FERPA (U.S. Department of Education, 1974) [USA]
- Personal Information Protection and Electronic Documents Act, or PIPEDA (Government of Canada, 2000) [Canada]

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Did you know?
In some countries, such as Germany, teachers can create safe learning spaces in FOBIZZ without needing to create personal accounts or share identifiable data. Such platforms are designed with education in mind and comply with strict data protection regulations.
Is there a similar alternative in your context?

Did you know?
Some AI platforms allow individual users to opt out of having their conversation used for model training through account level data control settings. However, this option must be activated manually, may not apply retroactively, and does not necessarily prevent temporary data retention for safety monitoring.

To illustrate, if you use ChatGPT, you can opt out of model training. Share these steps with students:

- Go to ChatGPT.com and log in.
- Click your **name or email** in the bottom-left corner to open the menu. Select **Settings**.
- In the Settings menu, click **Data Controls**.
- Look for the setting labelled **Improve the model for everyone** (or similar wording).
- **Toggle it off**. This prevents your data from being used to train and improve OpenAI models.

You can often find similar opt-out settings in other AI tools, though the labels and the scope of opt-out vary by platform.

A final thought
For now, at least, locally hosted or privacy-preserving AI systems are not always as practical, powerful, or user-friendly as many commercial cloud-based platforms. However, they do point to an important future direction for education: in principle, AI systems that run on a local device, an institutional server, or another controlled environment could allow teachers and institutions to make use of AI while reducing the risks associated with transferring learner data to external companies, governments or third-party platforms.

2.4. Principle II: AI-assisted language education should be responsible

Does our use of AI position our teaching as a force for positive change?

The imperative that the use of AI should foreground safety is complemented by the second principle, namely that the use of AI in language education should be responsible. Whereas the previous principle focused on the prevention of foreseeable harm and anchored ethical AI use to legal and institutional frameworks, the focus now shifts to producing positive change and connecting ethical AI use to the principles of democratic and sustainable futures. This involves the proactive deployment of artificial intelligence in ways that enrich the experience of learning and teaching languages, foster equitable learning opportunities, nurture linguistic diversity, promote social justice and encourage the ethical stewardship of social and environmental resources. The responsible use of AI can make language learning richer, more engaging and more rewarding for learners (see also Principle III).

This principle is expressed through two complementary guidelines: Guideline 3 (Promote social justice, linguistic and cultural diversity) focuses on the value-led aspects of AI-assisted language education. It takes as its starting points the awareness of structural injustices embedded in the AI technologies (e.g., whose languages and norms can be found in the training data), the ideologies that AI applications might reinforce, and inequalities that might exist in the ways that AI tools are accessed, and – aligning with a **critical** orientation to language education (e.g., Pennycook, 2021) – it deliberately challenges them. More importantly, Guideline 3 and the competences associated with it move practice beyond critique to a position of transformation, by seeking to understand how AI can be leveraged to create more equitable learning opportunities.

Guideline 4 (Foster environmental sustainability) focuses on ecological responsibility. The challenges it addresses relate to the environmental impact (e.g., carbon emissions, water consumption and energy use) of AI infrastructure, which includes data centres, model training and hardware production. Because this impact is rarely visible to end users, there is a risk that environmentally costly practices might be embedded in language education before their implications are fully understood. Like Guideline 3, this guideline adopts a transformative rather than merely cautionary orientation. It shifts the focus from recognising potential harms to supporting informed and intentional action, from classroom-level judgments about when AI use is pedagogically justified and in what manner (under what conditions and with what guardrails) and when low-tech alternatives may be sufficient, to institutional and policy-level decisions about accountability structures for AI procurement, implementation and use. This framing positions sustainability not as a narrowly technical or managerial concern, but as a multidimensional educational, ethical and governance challenge.

Did you know?

Even advanced large language models, which have been explicitly designed to reduce bias, may still exhibit bias implicitly. For example, they tend to associate negative language more often with Black individuals, they link women more frequently with the humanities rather than sciences, and they favour men in leadership contexts (Maslej et al., 2025).

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Guideline 3: Promote social justice, linguistic and cultural diversity

Just like other learning materials (Apple, 1992), AI systems encode structural hierarchies in language education. They reflect the priorities, blind spots and power structures of the contexts in which they were designed, and the languages, cultures and epistemologies that dominate their training data. In language education, where questions of identity, belonging and linguistic diversity are central, this is not an abstract concern. Guideline 3 therefore calls on language teachers to move beyond unreflective adoption of AI tools, towards a critically informed and transformative practice that actively uses AI as a lever for greater equity and inclusion.

Artificial intelligence holds both promise and risk for **linguistic diversity**. Power disparities between languages, linguistic varieties, and linguistic communities are well documented in the literature (e.g., Pennycook, 2021; Phillipson, 1992; Rosa, 2019). Artificial intelligence can counteract this potential bias by providing easier access to authentic or adapted materials in languages for which published learning resources may not be available. However, the uncritical use of AI in language education may amplify such disparities, as large language models (LLMs) tend to privilege languages (and varieties within languages) with high online visibility and status in AI output (Navigli et al., 2023), whereas under-resourced languages, including heritage languages, may be entirely absent. Challenging such bias might involve deliberately prompting AI tools in ways that surface marginalised languages, varieties and perspectives. It could also involve greater awareness of, and transparency about, the language varieties used as a model during accuracy-focused work, or a broader range of corrective alternatives.

AI tools do not enter language classrooms free of **cultural bias**: the datasets on which large language models are trained tend to reflect the norms of dominant social groups, which could exacerbate processes of cultural hegemony and epistemic injustice (Fricker, 2007; Resnik, 2025; Stracke, 2024). When this goes unrecognised, and language education is further reduced to instrumental goals, learners' identities and cultural capital risk marginalisation (Angelo, 2021; Atari et al., 2023; Helm et al., 2024). With appropriate training and resources, teachers who approach AI output critically, rather than as "objective" or "authoritative", are well placed to resist these tendencies, and to design AI-assisted activities that connect to established traditions of critical language pedagogy by actively affirming learners' cultural and linguistic repertoires (Jeon et al., 2025; Krulatz et al., 2018; Norton & Toohey, 2011).

Ensuring that AI-supported language education remains **socially just** also depends on the availability of technological infrastructure as well as the equitable and sustainable participation in digitally mediated learning environments, as described in the UN Sustainable Development Goals (United Nations, 2015). This involves addressing infrastructure barriers, such as disparities in access to electricity and high-speed internet (Bird et al., 2020; Broadband Commission for Sustainable Development, 2020), as well as barriers related to teachers' and learners' beliefs, digital confidence and opportunities for sustained training (Fuller, 1969; George et al., 2006; Venkatesh, 2022). To that end, developing the teachers' Technological Pedagogical Content Knowledge (TPACK; Mishra & Koehler, 2006) constitutes an important step that enables educators to exercise informed professional judgment, critically evaluate AI tools, and design contextually responsive pedagogies that mitigate inequity rather than reproduce it.

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Competences	Reflection questions
6. **[I can use AI applications in ways that sustain linguistic diversity]** *How effectively does my use of AI sustain or promote linguistic diversity?*	- *Which languages and varieties are represented in the AI tools I use, and which are absent or poorly served?* - *How do I ensure that feedback or resources reflect multiple linguistic varieties and registers?* - *When I use AI for accuracy-focused feedback, do I specify which language variety or register is the target, and do I share that information transparently with learners?* - *Is the AI output I use reproducing the same linguistic hierarchies I would challenge in a conventional textbook?* - *How might instances of linguistic bias in AI output be treated as pedagogical opportunities?*
7. **[I can mitigate algorithmic bias]** *How vigilantly do I ensure that AI output in my class does not reproduce biases towards certain knowledge, voices and values?*	- *Whose perspectives or interests are represented in the data that my AI tools rely on?* - *What prompting strategies might I use to actively generate an inclusive range of perspectives, knowledge systems or cultural framings?* - *What is my process for reviewing AI-generated content for bias before it reaches learners?* - *How do I create space for learners to bring their own knowledge and cultural perspectives into dialogue with AI output?*
8. **[I can ensure that the AI tasks I design are accessible to all learners]** *How do I make sure that the AI tasks I design are accessible to all learners?*	- *What disparities in access to AI tools am I aware of in my context?* - *How can I adapt my teaching to accommodate learners with limited access or skills?* - *What resources or partnerships can I use to bridge these gaps?*

Something to consider

This guidance is particularly relevant to language teachers working in Indigenous and community-based language revitalisation programmes. These settings, however, also raise additional considerations that fall outside the scope of this document. Some concerns include community sovereignty over language data, the near-total absence of such languages from LLM training corpora, and a heightened risk that uncritical AI use might normalise dominant norms where resisting them is the pedagogical purpose.

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Guideline 4: Foster environmental sustainability

Artificial intelligence has immense potential to transform language education, yet it also introduces a range of social and environmental responsibilities. The **environmental footprint** of AI tools, including carbon emissions, water usage and energy demands, has become a pressing concern (Luccioni et al., 2023). Building the necessary infrastructure, such as servers, computers and data centres, contributes heavily to resource consumption and environmental degradation, and the training of **large language models** also results in significant greenhouse gas emissions. Additionally, data centres generate substantial heat and require advanced cooling systems, leading to intensive water consumption.

Sustainable integration therefore requires calibrated use, rather than routine default use. At the level of classroom practice, educators can reduce unnecessary computation by designing **prompts** and workflows to maximise the quality of responses while minimising energy use. Another sustainable practice is to evaluate when and why to use AI. For example, when visuals are needed for teaching materials, existing image banks or open-access libraries may be preferable to prompting AI to generate new images. Similarly, for some information-seeking tasks, conventional web searches may entail substantially lower energy costs (10-30 times less energy) than AI-generated content (De Vries, 2023). At the institutional and provider levels, sustainability also depends on transparency and accountability. Encouraging AI vendors to disclose environmental metrics, such as carbon and water use per operation or per user, would enable educators and education systems to better understand and weigh the often-invisible externalities of AI deployment. This perspective is now endorsed across leading research communities in both environmental science and education.

One way to ensure that this potential is met would be to develop competences in effective prompting. The **IDEA framework** (Park & Choo, 2025) presented below helps guide this process, promoting clear and purposeful prompt design and iterative refinement.

Remember
Consider using digital content that is freely accessible to educators, in order to reduce environmental load.

Did you know?
...that (as of 2023) generating a single image with a powerful AI model like Stable Diffusion, DALL·E or Midjourney uses about as much energy as fully charging a modern smartphone. Less complex AI tasks, like text generation, require far less energy (Luccioni et al., 2023).

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Did you know?

Language teachers can use the IDEA framework (Park & Choo, 2025) to write effective prompts. This involves four steps: **I**ncluding essential parts, **D**eveloping with clear prompts, **E**valuating the output and refining prompts, and **A**pplying prompts with accountability.

- Including essential PARTS means that teachers should structure their prompts by identifying the following elements:
  - Person: an explicit statement of the expected AI role (e.g., "a teacher of X foreign language")
  - Aim: an explicit statement of the task, using an active verb (e.g., "design two learning activities")
  - Recipient: a specification of the target audience (e.g., "for learners aged X with language level Y")
  - Theme: context for the task, such as style, tone, constraints, and other relevant content (e.g., "design two learning activities about the structure of a narrative; use an example of a tale; include one good and one bad example; provide sentence models")
  - Structure: an indication of the desired output format (e.g., "in the form of a Word file", or "tasks should be accompanied by correct answers for self-checking").
- Developing with CLEAR prompts involves ensuring that prompts used to generate language learning activities or materials are Concise, Logical, Explicit, Adaptive and Restrictive or domain-specific.
- The third step, Evaluate and REFINE, requires **R**ephrasing keywords, and **E**xperimenting with context, using a **F**eedback loop to guide the generation of more accurate outputs. This is usually an **I**terative process, and as language teachers navigate through such iterations, they not only ensure the quality of the current interaction but also the future effectiveness of the system. Finally, **E**valuating and verifying the accuracy of responses is needed to ensure that learning outcomes and efficiency standards are met.

By applying these steps responsibly, perhaps using a clearly defined accountability checklist, language teachers can ensure that AI use in language education leads to improvements in the learning experience.

Competences	Reflection questions
9. **[I can understand the environmental consequences of using AI in language education]** *How well am I informed about the environmental consequences of AI use in language education?*	- *How do the tools I use affect people and ecosystems beyond my classroom?* - *What invisible costs (energy, data, labour) accompany apparently simple digital actions in my teaching practice?*
10. **[I can reduce the impact of AI use in my classroom]** *To what extent can I reduce the environmental impact of my AI use?*	- *When planning lessons, do I consider whether a low-impact or analogue alternative could achieve the same goal?* - *How do I plan, refine, and evaluate my prompts to maximise efficiency in AI use?*
11. **[I can promote environmental sustainability when using AI]** *How effectively do I teach and model environmentally sustainable uses of AI?*	- *How do I share examples of environmentally or socially responsible practice with colleagues?* - *What institutional policies or collaborations could reduce the environmental and social footprint of AI in education?*

More to explore

https://trails.upf.edu/

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2.5. Principle III: AI-assisted language education should be purposeful

How does AI add genuine pedagogical value and improve the quality of teaching and learning?

Ensuring that artificial intelligence is used in safe and responsible ways (Sections 2.3 and 2.4 respectively) provides a necessary backdrop for extending the thinking about the ethical use of artificial intelligence into the pedagogical core of language education. The third principle of the framework requires that artificial intelligence is deployed to support language education in ways that align with good pedagogical practice, and also that it enables activities that would not have been possible, or would have been prohibitively time-consuming, without the use of AI.

This principle is operationalised through two guidelines, as follows: Guideline 5 (Enhance theory- and policy-informed language pedagogy) focuses on pedagogical effectiveness. This involves encouraging the use of artificial intelligence in ways that build on effective teaching practices. What is pedagogically effective can vary depending on the perspective adopted and other contextual factors such as societal expectations, priorities in educational policy, available resources and more (Canagarajah, 2005; Kumaravadivelu, 2006; Pennycook, 2010). However, there is some consensus that effective teaching and learning are likely to share some features, such as a high degree of learner engagement, opportunities for differentiated instruction, sustained interaction with meaningful and authentic input, and opportunities for meaning-generation through authentic interaction (Ellis, 2003; Long, 2015; Philp & Duchesne, 2016), aligning with guidance outside language education about increased cognitive engagement (Chi & Wylie, 2014; Chi et al., 2018).

Complementing this, Guideline 6 (Create expanded language learning opportunities) underscores the transformative potential of artificial intelligence. The power of artificial intelligence lies not merely in its capacity to make teaching and learning routines more efficient; it also involves creating new learning affordances. What such novel affordances might look like in practice is still being discovered, but some directions are already visible. Artificial intelligence can, for instance, enable learners to engage with authentic materials in languages or varieties for which published resources are scarce, or to interact with texts and co-construct personalised learning trajectories in ways that respond dynamically to their interests and proficiency. Used in these ways, AI redefines what is possible in the language classroom (Puentedura, 2006).

The essential role of experimentation at this stage of adoption should also be acknowledged. Many educators are still exploring what these tools are capable of, and "purposeful use" will likely only emerge after a period of trial, error and discovery. In other words, the guidance here should not be seen as discouraging experimentation, which is not only inevitable but arguably desirable. However, as our collective understanding of AI's educational potential deepens, focus should increasingly shift towards intentional and effective application, ensuring that our use of these technologies remains purposeful whenever possible.

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Guideline 5: Enhance theory- and policy-informed language pedagogy

The effective use of artificial intelligence in language education is a function of the quality of uses rather than the quantity of AI tools and AI-supported activities. Quality, in this context, refers to how rich, inclusive and engaging language learning experiences are, and how artificial intelligence improves learning outcomes.

Effective language learning requires opportunities for **meaningful interaction**. This is an area where AI-powered conversational agents might prove helpful, by providing learners with low-stakes spaces for extended communicative practice, particularly in contexts where access to target language speakers is limited (Bibauw et al., 2019; see also Guideline 6). Such interactions can be designed to foreground meaning-making rather than formal accuracy, aligning with communicative and task-based approaches to language teaching (Ellis, 2003; Long, 2015). Additionally, AI can be used to efficiently generate materials for open-ended, contextualised tasks that require learners to negotiate meaning, express personal perspectives and engage in genuine communicative exchange, rather than mechanical practice. Using effective prompting (i.e., clear learning goals, defined outcomes and evaluation criteria; see also Section 2.4), such activities could draw on the learners' interests, experiences and linguistic repertoires, and provide necessary scaffolding to support the implementation of the tasks. Used in this way, artificial intelligence can be deployed to support the development of communicative competence, aligning with contemporary policy frameworks.

Additionally, AI tools can serve as valuable assistants in addressing class heterogeneity by supporting the **differentiation** of learning goals, content and tasks. They enable teachers to tailor materials to individual learners' levels, interests and needs, providing more personalised and inclusive instruction. For instance, speech-to-text functions can assist students who benefit from written transcriptions of spoken language, supporting learners with hearing difficulties or those who process information better visually. Conversely, text-to-speech functions can enhance language learning by introducing more spoken input, aiding pronunciation, listening comprehension and fluency. Teachers can generate multiple sets of exercises from the same input or adapt learning materials to match students' language proficiency levels. Language teachers can enhance language learning by developing multimodal learning supports assisted by AI, such as visuals, graphics, videos, interactive exercises and other multimedia resources.

Just as it has transformed classroom reality, the emergence of AI has radically changed **assessment** in language learning (Kherbouche, 2026). Traditional analogue assessment formats, such as essays and presentations, can no longer be unproblematically regarded as evidence of learning. AI challenges teachers to rethink evaluation practices, shifting the focus from artefacts of knowledge to the learning process, collaborative learning and the development of communicative competence. While mindful of legal restrictions on high-stakes assessment, teachers can leverage the ability of AI to provide learners with positive and constructive comments for feedback purposes in a number of ways, such as providing students with personalised feedback and **feedforward** based on transparent assessment criteria, or organising structured peer and group assessment. Furthermore, AI-assisted assessment can help mitigate some cases of unreliability in human assessment, such as teacher fatigue, timing (within the academic calendar or within a day), the examiner's physical or psychological state, the sequence of student submissions and subjective overreactions to certain error types, etc., which have been well documented before the emergence of AI (Fleming, 2007).

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Competences	Reflection questions
12. **[I can use AI in ways that foreground meaning-making and authenticity]** *How consistently do my AI-assisted practice opportunities foreground meaning-making and authentic language use?*	- *When I use conversational agents or AI-generated tasks for communicative practice, am I clear about what communicative goals I want learners to achieve?* - *When AI interaction prioritises formal accuracy over meaningful communication, how do I notice this, and what do I do about it?* - *How do I compare the quality of communicative engagement in AI-supported activities with that in teacher-led or peer interaction tasks?*
13. **[I can design clearly structured language learning tasks with the help of AI]** *To what extent do the tasks I design with AI support have clear learning goals, defined communicative outcomes and transparent evaluation criteria?*	- *How do I ensure that the outcomes of AI-supported communicative tasks are defined broadly enough to allow for genuine language use, rather than channelling learners towards a single expected response?* - *When designing AI-supported tasks, can I co-create prompts with learners to draw on their interests, experiences and linguistic repertoires?* - *When an AI-supported communicative task does not produce the learning outcomes I intended, what does my process for reviewing and revising the task look like?*
14. **[I can personalise language learning experiences through the use of AI]** *How successfully can I use AI tools to personalise learning?*	- *Which elements of my teaching (e.g., goals, content, tasks, feedback) am I currently differentiating with AI?* - *How might I use speech-to-text, text-to-speech or other assistive AI functions to support learners with specific needs without singling them out?* - *How do I evaluate whether AI-assisted differentiation is actually improving learning outcomes for the learners it is intended to support, rather than simply generating more varied materials?*
15. **[I can use AI to help provide feedback in responsible ways]** *How transparent and effective is the way I integrate AI into feedback processes?*	- *When I use AI to generate feedback for learners, how do I communicate to them that AI has been involved?* - *When AI feedback and my own professional judgment diverge, how do I handle that divergence transparently with learners?* - *How do I know whether AI-assisted feedback is actually improving learner performance and self-regulation over time, rather than simply providing more feedback more quickly?*

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Guideline 6: Create expanded language learning opportunities

The question of whether artificial intelligence adds value to language teaching and learning requires a qualitative judgment of what AI use makes possible that was either impossible or pragmatically impractical previously. A useful starting point in this judgment is whether AI use replicates existing practice or genuinely expands what teachers and learners can do. What counts as desirable change is context-dependent and ideologically contested (Alim & Paris, 2017; Hornberger & Johnson, 2007; Pennycook, 2010), so Guideline 6 does not focus on examples of promising use; rather, it explores how the intentional use of artificial intelligence can create affordances that expand learner opportunity, enhance learner agency and increase the teachers' capacity. These criteria are not a checklist: they can be applied together or separately depending on the context and goals of language education in any specific setting.

Expanding **learner opportunity** involves the use of artificial intelligence to generate learning affordances that were previously unavailable or prohibitively resource-intensive. Examples of such affordances include providing opportunities for communicative practice in languages for which human talking partners are scarce, and generating personalised feedback at a scale that would be impractical for most teachers. The key diagnostic is that the use of AI in such cases involves the transformation of the language learning experience, as opposed to the superficial or tokenistic reproduction, in a faster and less costly format, of activities that already existed, such as drill exercises and closed-form questions. The speed with which generative AI can generate drill-type items (for example, vocabulary practice or multiple-choice questions) should be weighed against their variable language-learning potential, ensuring that such exercises serve clear pedagogical purposes and complement, rather than replace, higher-involvement and meaning-focused work.

A second criterion is the impact of artificial intelligence on **learner agency**, i.e., their capacity to direct, shape and take ownership of their learning. The transformative use of artificial intelligence involves practices and resources that support learners in setting their own learning goals, evaluating their own output and making meaningful choices about how and what to learn. This is to be contrasted with usages of artificial intelligence which constrain agency, either by algorithmically channelling learners through pre-structured learning pathways or by foregrounding easily measurable outcomes. What differentiates transformative use from mere automation is the degree of control that learners enjoy over their learning goals, process and outcomes. This is especially important for language education, which is inextricably linked to considerations of identity, motivation and willingness to communicate, all of which may be impacted by the use of AI (Fathi et al., 2024; Liu & Reinders, 2025; Yuan & Liu, 2025).

Turning to the teachers, the third criterion for the intentional transformation of the language learning experience concerns what AI makes possible for teachers. When thinking of **teacher capacity**, the focus is not just on time saved but on the expansion of their pedagogical horizons. Automation of well-specified, low-judgement tasks (e.g., first-pass feedback, routine differentiation) is valuable insofar as it frees teacher time for work that is difficult to automate responsibly: interpreting learner trajectories, building classroom relationships and orchestrating dialogic interaction. However, the use of AI to merely replicate what teachers already did does not genuinely expand capacity in the

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sense discussed here. AI becomes valuable when it helps teachers notice what they would otherwise miss, explore new pedagogical approaches or design learning experiences that would otherwise have been beyond their reach. Transformative education, in this sense, involves making teachers more capable and more thoughtful practitioners.

Did you know?

AI can help achieve learning goals that would otherwise be out of reach. For example:

- **Richer feedback loops**: AI can provide learners with a first layer of feedback on written or spoken output. This can free teachers to focus on deeper, teacher-led analysis. Subject to legal constraints (see Guideline 1), such a hybrid division of labour can drive a genuine refocusing of language lessons, away from recognition and functional literacy, and towards **critical** literacy and metalinguistic insight.
- **Challenging normative monolingualism**: AI can be leveraged to push back against the monolingual norm in language classrooms by building connections to the full linguistic repertoires of plurilingual learners. One way to do this would be to translate content to and from the learners' other languages, which would have been impractical for teachers in very diverse classrooms.
- **More cross-lingual work**: The use of AI in language education enables scaffolding that leverages the students' linguistic repertoires, which would have been impossible in the past if the teacher did not speak or understand the L1s of the students.
- **Learner-centred syllabus design**: AI can support a shift from top-down textbook-based instruction to context-sensitive, bottom-up syllabus design. This can generate customised materials based on learners' strengths, needs, and interests.

Did you know?

The Substitution, Augmentation, Modification or Redefinition (SAMR) framework (Puentedura, 2006) describes digital innovation as leading to one of four outcomes:

- When technology **substitutes** an existing task, there is no functional change. This can happen when an AI application writes multiple-choice tasks for grammar practice.
- When technology **augments** an existing task, there is some functional improvement, but the nature of the task remains the same. A teacher using AI to generate multiple versions of the same grammar exercise is still operating within the traditional paradigm, but AI makes differentiation faster and more scalable.
- When the new technology allows for the substantial redesign of an existing task, this leads to **modification**. An instance of this might be when learners use a conversational agent that dynamically adjusts to their interest and linguistic profile, rather than using a scripted dialogue.
- When the new technology enables the creation of new tasks that were inconceivable without it, this leads to task **redefinition**. This could involve working with AI tools to co-construct individualised learning trajectories by selecting authentic texts (e.g., podcasts, news articles) suited to the learners' level and interests, and having the AI application generate vocabulary support, comprehension questions, and follow-up tasks.

Competences	Reflection questions
16. **[I can use AI to transform language learning experiences]** *To what extent do the AI-assisted tasks I design transform the language learning experience?*	- *What is one example of a learning opportunity you created using AI that would otherwise not be feasible in your classroom or learning context? In what ways have students benefited from this?* - *What would be lost if I removed AI from a task entirely?* - *How do the AI-supported activities I use complement meaning-focused work? What pedagogical purposes do they serve beyond efficiency?* - *What are the learning outcomes that my AI-supported activities serve, and how well do they align with my instructional goals?*
17. **[I can use AI in ways that support the agency of language learners]** *To what extent do the AI-assisted tasks I design increase learner agency?*	- *Do the AI tools I use open up choices for learners, or do they channel them through pre-structured pathways?* - *How much control do learners have over their goals, process and outcomes?* - *How might the AI tools I use affect learners' sense of ownership over their language learning?*
18. **[I can improve my teacher capacity through the use of AI]** *To what extent do the AI-assisted tasks I design improve my teacher capacity?*	- *How am I using the time that is saved by delegating aspects of my work to artificial intelligence?* - *In what ways might AI help me notice things about my learners or my practice that I would otherwise have missed?* - *Which of the AI tools I use are helping me do new things as a language teacher, and which ones are helping me do the same things more efficiently?*

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2.6. Principle IV: AI-assisted language education should be human-centred

How can teachers and learners use AI reflectively and maintain professional agency?

A final principle in the framework concerns the centrality of human agency in AI-assisted language education. At its heart, this principle affirms that the introduction of artificial intelligence into the language classroom should strengthen the capacity of both learners and teachers to engage with learning and with language (Séror, 2026), including AI-generated language, as informed, purposeful and critically aware agents. For learners, this involves developing the competences needed to use AI tools thoughtfully and ethically, while preserving their voice, identity, and plurilingual repertoire. For teachers, this agency is understood as "socially constituted capacity to act" (Barker, 2008, p. 234) in an ever-changing ecology of shifting policies, curricula, technological affordances and developmental opportunities. Artificial intelligence, when thoughtfully deployed, can enrich both dimensions of this ecology.

This principle is operationalised through two complementary guidelines. Guideline 7 (*Prepare learners for critical, healthy engagement with AI*) focuses on the learner as a developing agent in an increasingly AI-saturated world. It draws on the well-established traditions of critical literacy in language education to help learners relate to AI-generated content in a thoughtful way, while also attending to their well-being and protecting the authenticity of their voice and plurilingual identity. Crucially, this guideline positions AI literacy not as a separate technical competence but as something that can be developed through, and in service of, language learning itself.

Guideline 8 (*Empower language teachers as reflective professionals*) complements this by positioning teachers as inquirers into their own practice. Drawing on established frameworks for reflective practice (Farrell, 2022; Schön, 1983), it explores how artificial intelligence can support reflection, alongside more sustained forms of classroom-based inquiry. Here, AI serves not as an authority, but as a scaffold: a tool that can help teachers surface assumptions, interrogate their practice, and engage in evidence-informed professional dialogue.

Together, these guidelines expand the horizon of pedagogical and professional possibility for both teachers and learners. The human-centred use of artificial intelligence creates new affordances for action (Stelma & Kostoulas, 2021), empowering teachers to respond to classroom challenges, experiment with emerging pedagogies, and negotiate the institutional expectations that shape their work – while simultaneously equipping learners to engage critically and creatively with the AI tools that are increasingly present in their lives. Such empowered, reflective engagement has the potential to gradually reshape the language education ecology itself, aligning practice more closely with the broader commitments to equity, plurilingualism, and human dignity that underpin this framework.

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Guideline 7: Prepare learners for critical, healthy engagement with AI

Artificial intelligence is increasingly present in the lives of language learners, both inside and beyond the classroom, often without critical framing that would ensure its effective and safe use. This guideline examines how language education can provide learners with competences needed to engage with AI tools and AI output in a thoughtful, ethical and effective way. This involves developing AI literacy skills, ensuring learner well-being in learning environments where human and non-human input co-exist, and preserving the learners' identity, voice and plurilingual repertoire in spite of the dominant/standardised norms and content that **LLMs** tend to (re)produce.

Language education is well positioned to foster the competences that learners need to develop **AI literacy**, especially when working in traditions that foster critical literacy (Edge, 2006; Pennycook, 2010). Evaluating texts for accuracy, register, communicative and cultural appropriateness and ideological stance are some of the practices that are likely to exist in language learning curricula, and these can be readily leveraged to compare human- and AI-generated texts. Language teachers can rely on established disciplinary strengths to help learners understand how discourse is generated, and classroom work can extend the collaborative critique of hallucinations and embedded bias. Similarly, existing practices in teaching speaking and writing can be deployed to develop effective prompting strategies, and to interrogate, revise, and respond to AI output. Such activities can simultaneously function as language learning and AI literacy education.

To ensure **learner well-being**, teachers can take steps to calibrate AI-supported tasks and practice opportunities with authentic human interaction. Among other things, they can monitor for isolation, fatigue and disengagement associated with excessive screen time or AI overuse (Deep & Chen, 2025; Lepp et al., 2022; Zhang & Liu, 2025). An additional concern is that digital routines in the language classroom should remain age-appropriate and consistent with the learners' cognitive strengths and needs.

A third concern relates to **learner identity**, and the risk that this may be eroded by the increase in AI-supported practices. Language education has a long humanist tradition, with a commitment to voice, personal narrative and respect for diversity that can be usefully brought to bear here. For example, teachers may design AI-supported tasks that use learner experience and personal narrative as the starting point, treating AI-generated content as a foil or talking partner rather than a model, and inviting learners to compare, contest and personalise its output. Another possibility involves using AI-supported speaking and writing tasks in ways that preserve space for genuine authorial choice, by framing AI as a resource for drafting or editing rather than a source of finished text. Additionally, it is possible to leverage the multilingual capabilities of AI to create space for dynamic cross-linguistic practice, using translation, comparison and code-switching as resources for meaning-making.

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Competences	Reflection questions
19. **[I can use AI to support the development of (critical) literacy skills]** *How well do I integrate language practice and AI literacy development in my teaching practice?*	- *Which critical literacy practices in my current teaching can I readily transfer to the evaluation of AI-generated texts?* - *How do I help learners critique AI-generated content, and what language learning opportunities does this create?* - *What can I do to integrate AI literacy work in meaningful language learning activity rather than treat it as a separate technical skill?* - *How might effective prompting be taught as a language learning activity (e.g., by connecting to existing work on register, genre, and communicative purpose)?*
20. **[I can use AI to nurture learners' well-being]** *How effectively do I preserve learner well-being in an AI-assisted environment?*	- *What signs of fatigue, disengagement, or over-reliance on AI do I look for during AI-supported activities? How do I respond when I notice them?* - *How do I adapt AI-supported tasks to accommodate learners with different cognitive strengths, learning differences, or levels of digital confidence?* - *What can I do to encourage learners to reflect on their own experience of AI-assisted learning, its effect on their well-being and possible over-dependence?*
21. **[I can use AI to support the development of the learners' identity]** *How well do the AI-assisted tasks that I design affirm learner identity?*	- *What are my teaching options, when AI-supported tasks appear to be producing uniform or formulaic output that overrides learner voice?* - *In what ways do I invite learners to draw on their own experiences and perspectives to compare, contest, or personalise AI-generated content?* - *What procedures do I have in place to help learners notice their authorial choice, tone, stance, and personal expression and compare them to AI output?*

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Guideline 8: Empower language teachers as reflective professionals

Language teachers working in AI-assisted classrooms must simultaneously manage the immediate demands of instruction and maintain a reflective distance from their practice. Artificial intelligence, used thoughtfully, can support both of these demands by helping teachers plan more deliberately, observe more systematically, and interpret their teaching more deeply. Guideline 8 draws on established frameworks for reflective practice (Farrell, 2022; Schön, 1983) to propose ways in which AI can support teachers as inquirers into their own language teaching.

Just like any other instance of teaching, AI-assisted language education also requires that teachers engage in regular **reflection** about their professional practice. This involves being able to effectively plan forward (reflection-for-action), which can be supported, for example, by generating variants of a lesson or activity with different degrees of autonomy (e.g., tightly scaffolded vs. open-ended), or counterfactual scenarios (e.g., "what would happen if I only had one student?") and making structured comparisons between them. It also involves the ability to make pedagogical judgments in (almost) real time, by deploying keen observation skills and pedagogical sensitivity for monitoring the implementation of AI-assisted activities. Finally, **conversational agents** can be leveraged to provide input and feedback after lessons or longer instructional sequences, prompting reflection on the less "accessible aspects of teachers' work" (D. Freeman, 2016, p. 208), and thus making professional action more purposeful.

Artificial intelligence can also be used to support forms of **professional development**, such as **classroom-based inquiry**, whether framed as action research (Burns, 2010), exploratory practice (Hanks, 2017), or teacher-led scholarship (Sato & Loewen, 2022). For example, teachers can assign **large language models** the role of a critical friend whose task it is to help define and collaboratively refine a teaching-related puzzle. Alternatively, they could reformulate broad areas of investigation into more precise and researchable research questions. Assuming ethical and legal safeguards are met (see Section 2.3), artificial intelligence can also be used to transcribe audio or video recordings of classroom interactions, identify recurring patterns such as specific grammatical deviations or discourse markers, or compare student outputs or proficiency across lessons or groups, thus facilitating systematic analysis.

With appropriate prompting, AI can promote **teacher reflexivity** (Edge, 2011) by helping teachers theoretically position themselves and their teaching practices. Choices about how AI is used inevitably position language education within a broader landscape of values. For example, a teacher who routinely uses AI to generate multiple-choice grammar items may be communicating to learners that language learning is mostly about accuracy and measurable outcomes. By contrast, a teacher who uses the same tools to scaffold creative writing might be making different statements about what language education is for. Using AI tools to interrogate and make explicit the pedagogical assumptions embedded in a lesson plan or activity, or to propose an alternative version grounded in a different set of values, can make these choices visible and deliberate rather than habitual. Over time, such reflective habits shape how learners, school leaders, policymakers, and the public come to understand language teaching as an intellectually serious and socially committed professional practice.

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For example...

Here's a prompt that can be used to reflect on a lesson

"You will assume the role of a critical friend and experienced language educator. Your task is to help me reflect on a lesson I taught today to upper-intermediate learners on expressing opinions in English. The main activity was a debate, and I integrated an AI chatbot for brainstorming arguments. I want to evaluate the effectiveness of this lesson in supporting learner autonomy and engagement. Ask me probing questions about the structure of the lesson, the role of AI, and learner responses. Please focus on: (a) what worked well and why, (b) what I need to do differently in the future, and (c) how AI influenced the dynamics of the lesson. Present your questions in a way that encourages me to reflect critically and constructively, so I can improve the next iteration of this lesson."

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Competences	Reflection questions
22. **[I can use AI to strengthen my reflective practice]** *How rigorously do I use AI to reflect on my teaching practices?*	- *How often do I use AI tools to revisit a lesson or activity?* - *When I ask an AI tool to comment on my teaching, do I provide enough contextual detail for its feedback to be genuinely useful?* - *How well do the patterns an AI tool identifies in my practice align with my own professional interpretations?* - *How might I systematically record AI-assisted reflections over time to track the development of my thinking?*
23. **[I can critically reflect on my use of AI]** *How critically do I use AI to systematically explore aspects of my teaching practice?*	- *How do I identify which aspects of my teaching are worth investigating?* - *Which aspects of classroom-based inquiry can be handled well by artificial intelligence and which require more human engagement?* - *Do I use AI tools to surface assumptions in my lesson plans or materials that I might not have noticed myself, or do I mainly use them to confirm decisions I have already made?*
24. **[I can align my use of AI with my teaching values]** *How deeply can I reflect on the values implicit in my use of artificial intelligence?*	- *Thinking about the AI-supported activities I use most frequently, what do they collectively suggest about my teaching beliefs?* - *How would I explain the values embedded in my AI use to a learner, a parent, or a policymaker?*

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Part 3

Usage cases



3. Usage cases

This chapter of the Guidelines provides realistic examples of how AI might be used in language education, accompanied by commentary grounded in the framework. These examples are provided to illustrate the critical evaluation that the guidelines can scaffold. They are not intended as models of "best use", and – similarly – the commentary reflects one way of thinking about the examples.

Case 1: *Passé composé* or *imparfait*? A grammar worksheet with an AI makeover

Overview of the lesson

A B1 French as a foreign language class in lower secondary education. The class is working on the contrast between the *passé composé* and the *imparfait*, which has already been presented in class as the distinction between completed events and background descriptions or habits. To help the students practice, the teacher uses a general-purpose conversational agent to write 20 gap-fill sentences contrasting the tenses, as well as an answer key. She then asks for the sentences to be rewritten as a short story about a group of teenagers on a school trip, and uses an image generator to create cartoon illustrations that scaffold the task. The learners complete the worksheet individually on paper, checking their answers in pairs against the AI-generated answer key, and are allowed to ask the conversational agent for clarifications, in their L1, if they are not comfortable asking the teacher. The lesson concludes with a brief explanation of the most frequent errors. The teacher estimates that preparing the worksheet took about 10 minutes, compared with the 40 minutes it would normally take.

Discussion

Safe use	Responsible use
On the whole, this is a low-risk use of AI. The use of the AI by the learners is very limited in time and scope, and assuming that there are no age-related legal constraints preventing the use of AI, there are few concerns relating to **Guideline 2**. Regarding legal ramifications (**Guideline 1**) it is likely that the AI-generated images will have to be explicitly labelled as such. A check against the image tool's terms on educational and commercial reuse may also be required, as some services require premium licenses for professional use. This is especially relevant if the worksheet is later shared with colleagues or uploaded to a school platform.	The answer key raises concerns about the teacher's accountability. The *passé composé* and the *imparfait* are often both grammatically possible, marking differences in meaning. The key therefore needs the same review as any material the teacher would write. The story framing raises questions about the stereotypes being modelled, which are relevant to **Guideline 3**. It is likely that the AI's default description of a school trip is set in Paris. Specifically asking for a setting elsewhere in the francophone world (e.g., Marseille, Brussels, Montréal or Yaoundé) is a minimal modification which can nevertheless challenge the assumption that French exclusively belongs to France. The environmental impact (**Guideline 4**) of text generation is trivial, but AI-generated images have a substantial

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	ecological footprint, which is difficult to justify if the image is mostly decorative. The teacher will need to decide whether the image provides necessary scaffolding, and whether an open-access image bank, or a hand drawing, might serve the same purpose.
Purposeful use	Human-centred use
The conservative outlook of this activity need not be seen as criticism from the perspective of **Guideline 5**: there is value in controlled practice when learning a language, and reducing teacher preparation time is also useful. However, there is scope for considerable improvement as regards expanding pedagogical possibility (**Guideline 6**). In terms of the SAMR framework (p. 42), this lesson is a typical example of substitution (the AI produces a worksheet that the teacher could have written, and the story frame and illustration do not change what learners do with it). This could move towards augmentation or modification with only small modifications. For example, the teacher could generate versions at two levels of difficulty, and learners could be asked to supply experiences of their own for the AI to turn into practice items. Such changes could support higher-involvement work rather than simply reproduce a drill more efficiently.	This lesson does very little to develop AI literacy (**Guideline 7**). This could be achieved relatively simply by informing the learners that the sentences and key were AI-generated and inviting them to consider where the key may be wrong. The case also lends itself to the reflexive questions in **Guideline 8**. A teacher who routinely uses AI to produce grammar drills may be signalling, without intending to, that language learning is mainly about accuracy and measurable outcomes. Reviewing which AI-supported activities one uses most often, and what they suggest about one's beliefs, may be a good starting point for such reflection.

Overall assessment

This case is useful precisely because it is ordinary. It probably represents much of how AI is used in language classrooms: quickly, by the teacher, to speed up preparation of familiar materials. It is safe, efficient and broadly unobjectionable, and the framework does not suggest it should be avoided. What the framework does is change the questions asked about it. The issue is less whether the worksheet is acceptable than whether AI has added anything beyond speed and decoration, and the answer is: not yet. The refinements suggested above are small: reviewing the key, deciding which variety is the target, dropping a costly decorative image, and using the time saved for a meaning-focused follow-up. Together they would move the lesson from substitution to a use of AI that makes a pedagogical difference.

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Case 2: How not to do it

Overview of the lesson

An A1–A2 class learning Estonian is reading *Sipsik*, a classic Estonian children's book by Eno Raud, illustrated by Edgar Valter (1962), about a rag doll made by a boy as a birthday present for his sister, which comes to life and shares the children's adventures. The language objectives are for learners to describe a character's physical appearance and personality, to formulate and answer questions about a literary text, and to use basic descriptive vocabulary and sentence structures. Because some learners cannot yet read the original, the teacher uploads the entire book to a general-purpose chatbot and asks it to simplify the text to A1–A2. After reading, learners ask and answer questions about the story and describe Sipsik in their own words. Next, they describe the doll to an image generator, gradually refining their prompts, and produce a series of images that match their descriptions.

Discussion

Safe use	Responsible use
While the teacher's rationale for simplifying challenging linguistic input makes sense for differentiation, uploading an entire book that is still under copyright is hard to justify under **Guideline 1**. A synopsis or a short extract might be acceptable alternatives. The images, which reproduce a protected character and illustration style, may also be legally problematic. As for learners using the image generator directly, this requires that they meet the tool's age requirements and terms of service. At minimum, teachers should familiarise learners with data-control settings and, where possible, switch off the use of their data for training (**Guideline 2**). Where available, a licensed educational workspace, a purpose-built tool or a tool configured not to use inputs for training would be a preferable option.	Generating a series of images per learner comes at a substantial energy cost, which is difficult to reconcile with **Guideline 4**, especially since this activity does not seem to correspond to valuable learning objectives (see also the discussion under "purposeful use"). With regard to **Guideline 3**, the activity also raises some questions: Sipsik's appearance is fixed by Valter's illustrations, which are part of a shared cultural heritage, and a generic image may be unable to match it. This creates an opportunity to compare the generated images against the original ones, which may be pedagogically desirable.
Purposeful use	**Human-centred use**
The language tasks in this lesson (understanding the story, formulating and answering questions, describing a person using new vocabulary) are cognitively engaging. However, the image-generation stage adds little to this (**Guideline 5**). Per **Guideline 6**, two key considerations, for this and any similar task, are (a) how generating an image	**Guideline 7** requires that learners keep the main decisions, such as their interpretation of the story, their description of Sipsik and their judgment of what the AI produces. A closing reflection may highlight the role of human agency (e.g., *What did we learn by using AI? Which details did the AI misunderstand or leave out? Why is the*

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helps learners learn about describing, comparing or revising their text, and (b) to what extent might this be possible without the use of artificial intelligence.	*AI image different from Edgar Valter's illustration? Who should decide what Sipsik looks like: the author, the illustrator, the AI or readers?*) For the teacher (Guideline 8), the case is a reminder to ask about each AI step whether it serves an objective or only fills time.

Overall assessment

What this case shows is that even well-designed language activities may lose focus by including AI steps that add nothing beyond performative value or unfocused entertainment. In the original lesson, every step (e.g., simplifying a text for weaker readers, letting learners "see" their descriptions) seems pedagogically reasonable. However, the way in which they are operationalised creates a range of copyright exposure issues, an avoidable environmental cost, and a task that keeps learners busy without challenging them. This does not suggest that artificial intelligence is harmful. Rather, it serves as a reminder for teachers to start their planning from the learning objectives and consistently ask themselves how each use of AI contributes to achieving them.

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Case 3: One text, three versions: AI-supported differentiation in a mixed class

Overview of the lesson

Three new learners, who have A1 level of linguistic proficiency in German, have arrived in a mixed-ability German as a second language class in a lower secondary school, where most learners are at A2–B1, although two children have a diagnosed reading difficulty. The class is about to start a unit that contains a B1 level text about plastic waste. The teacher uses an AI application to produce three parallel versions of the text at A1, A2 and B1 levels, retaining the same key information. The A1 version contains a glossary of difficult words translated into the three newcomers' home languages (Arabic, Ukrainian and Tamazight), and the A2 version contains a glossary with simple definitions in German. To decide which learner receives which version, he pastes a spreadsheet with learners' names, previous test scores and notes on reading difficulties into the agent and asks it to suggest groups. In class, learners read their assigned version, complete level-appropriate comprehension questions, and then work in mixed-level groups to produce a poster about reducing plastic use at their school.

Discussion

Safe use	Responsible use
From a safety perspective (Guideline 2), the grouping step seems quite problematic. The information in the spreadsheet (e.g., names, assessment results and information about reading difficulties) is personal data of minors, some of which is sensitive. In addition to the legal ramifications (Guideline 1), pasting this information into the AI system is a form of profiling, since artificial intelligence is used to sort learners on the basis of incomplete data. Such a step is hard to justify, as the teacher presumably already knows the learners and should be able to assign learners to groups more quickly and accurately without AI. The adaptation of the textbook text raises a smaller question concerning copyright law (Guideline 1). Before modifying the text, the teacher would need to make sure that such adaptations are permitted.	The home-language glossaries are a particularly interesting feature of the case from the perspective of Guideline 3, as they expand the linguistic diversity of the schoolscape. In addition to legitimising less visible languages, these glossaries build bridges to the newcomers' existing linguistic repertoires, in ways that would be highly impractical without AI. One caveat to bear in mind, however, is the uneven quality of AI translations, especially when it comes to poorly documented languages such as Tamazight. Although the teacher might not be able to verify the translations, this might be done by the learners themselves or their families, and involving them would also valorise their expertise. With regard to Guideline 4, generating three texts is a modest cost for a resource that can be reused.
Purposeful use	**Human-centred use**
This case illustrates the kind of differentiation described in Guideline 5: the same content adapted to different levels, with audio support for learners who process information better by	As long as the glossaries are presented as a resource for everyone rather than as support for 'weak' students, this lesson is helpful in nurturing the learners' home

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listening. Considering how time-consuming and resource-intensive such adaptations are, the use of AI produces a substantial gain. Care should be taken, however, that simplification does not remove language structures or content that are supposed to be taught. The teacher should check that the three versions keep the same key ideas, as the prompt intended. As regards creating new learning possibilities (**Guideline 6**), the availability of multiple versions of the text creates the opportunity for learners to exercise some autonomy, if they are allowed to select the materials that they will use. This turns differentiation from something done to learners into something they control.	languages and identities, as per **Guideline 7.** For teacher development (**Guideline 8**), the key reflective question connects to the decisions made concerning differentiation. The teacher needs to reflect on whether AI-supported differentiation is actually improving learning for the learners or is simply generating more varied materials. Comparing the posters and the comprehension answers across levels can provide the teacher with some insights into how the task works.

Overall assessment

This case demonstrates how artificial intelligence can be leveraged to make a meaningful difference for inclusion. This is achieved by making a lesson accessible to learners at very different levels, in several languages and modes, at a sustainable cost in preparation time. While the overall design (learners working on differentiated input and then collaborating in a shared task) is pedagogically sound, the case also shows how well-intentioned differentiation can easily create risks of its own (compromising sensitive data, oversimplifying content, labelling learners). Such risks can be mitigated by principled decision-making regarding who decides and who sees what: the groups should be formed by the teacher, rather than the AI; the learners can be offered choice in selecting their version; and learners and families can be involved in validating the home-language glossaries. Such relatively simple changes can make the lesson more inclusive.

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Case 4: Who does what at home? Gender roles in an AI-generated phrase bank

Overview of the lesson

An A1–A2 English class of learners aged around eleven is working on a lesson focusing on family members vocabulary and structures connected to their daily routines. The lesson, which is is framed as a small inquiry (*Let's find out what our family members do all day to keep family life running smoothly*), has a dual outcome. In terms of language, learners should be able to describe everyday activities in their families, using the present simple and adverbs of frequency. In terms of content, they are expected to critically analyse AI-generated outputs by identifying gender stereotypes. Before the lesson, the teacher enters the same prompt into two or three conversational agents (e.g., Le Chat): "Make a phrase bank of 20 sentences for an English A1–A2 class about family members and their daily routines. Use an adverb of frequency (always, usually, often, sometimes, rarely, never) in each sentence." The outputs include sentences such as *My mother usually cooks dinner in the evening*, *My father usually watches TV after work* and *My brother often plays video games in his free time*, and the teacher prints them as a worksheet. In class, learners first work individually: they tick the sentences that are true for their family and correct the ones that are not (*My father rarely watches TV after work*, or *My father usually walks our dog after work*). They then compare their versions with a partner (*Who prepares breakfast in your family? Usually my mother does, but sometimes my father does*) and record how well the list matched their family (0–5, 6–10, 11–15 or 16–20 sentences). A teacher-led discussion follows, comparing the lists produced by the different tools: whether the sentences describe the families learners know, which tasks are given to women or men, parents or children, and who decides what happens at home. A final step connects the discussion to learners' own lives (*What roles do you have in your family now? What can change when you become a parent? Which roles stay the same?*) and leads to a short descriptive paragraph.

Discussion

Safe use	Responsible use
The main compliance question here is age, as students must meet the minimum age requirement for interacting directly with generative AI tools, but in many settings English learners at A1–A2 level may be below it (this may be less relevant for other languages where instruction begins later). In many jurisdictions, children under 13 might be restricted from interacting with AI. Additionally, the tool's terms of service and institutional policy need to be checked before the lesson (**Guideline 1**). To avoid such complications, learners work with printed outputs rather than interact with AI tools. In variations of this example, where when students meet the age requirement and their interaction with AI tools is	The learning activities in the lesson are designed to encourage critical analysis of the generated examples, and reflection on the social roles attributed to different genders, age groups and social classes, all of which align well with **Guideline 3**. Such activities encourage students to recognise how role-allocations may be stereotypically feminised or masculinised. The discussion, however, requires particular care, as gender roles in learners' families are shaped by cultural and religious norms, and some learners may hold them strongly. The aim is for learners to notice that roles differ across families and that the AI presents one pattern as the default without passing judgment on any family's

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desirable, they should be asked to select safe tools (e.g., DuckAI), review the data-control settings and, where possible, disable the use of their data for model training (Guideline 2). Group work (in addition to its pedagogical advantages) offers a second safeguard, as fewer individual accounts are required. The discussion of the learners' own families should ideally take place among the learners, keeping personal information offline.	arrangements. Language-wise, if this discussion extends beyond the learners' A1–A2 linguistic resources, the teacher may need to decide how much can take place in the learners' other languages. Regarding Guideline 4, the environmental impact of AI use can be reduced by having students work in groups, thus limiting individual interactions with the tool and reducing the energy consumption, carbon footprint and water usage associated with data centres.
Purposeful use	Human-centred use
As language practice, the AI's contribution is modest. A teacher could write a similar list, and the adverb-of-frequency requirement makes the target structures visible. The pedagogical value of the task could be sharpened, per Guideline 5, by making the linguistic objectives explicit: the present simple for routines, adverbs and expressions of frequency (*usually*, *on weekends*) and *like* + infinitive. The purpose of the activity, however, extends beyond conversational practice and learning vocabulary and syntactic structures. Comparing AI output across groups can make the constructedness of the output visible. The bias in the output is leveraged as content of the lesson to help develop critical digital and AI literacy, in ways that would not have been possible without AI (Guideline 6).	The reflective activity following the use of AI encourages students to make informed decisions, reflect on their language choices and recognise the social biases reproduced in AI-generated content and language more generally. The closing questions (*What roles do you have in your family now? What can change when you become a parent?*) are a good example of Guideline 7 in practice, since they use the learners' own experience as the starting point. As regards Guideline 8, the phrase banks and learners' own descriptions of their families together give the teacher material for reflecting on whether the lesson changed the learners' perceptions of who does what at home.

Overall assessment

This case illustrates how a short, low-tech use of a conversational agent can serve two outcomes at once. Its strength lies in turning a predictable weakness of AI output, the reproduction of social stereotypes, into the object of study. While phrase banks have been extensively used in language learning, the lesson extends beyond conventional language learning to develop critical awareness about biases encoded in AI and language more generally. Some refinements, as outlined in the discussion above, may help by making implicit aspects of the lesson more explicit, and demonstrate how improvements can be made to ensure safer and more purposeful use and show that the AI does not need to be in learners' hands for the lesson to develop critical AI literacy.

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Case 5: Interviews with people from the Commonwealth

Overview of the lesson

A teacher is teaching a B1–B2 CLIL class integrating English and History. They have already covered the Stuart era, including the Civil War and the Commonwealth. Rather than assign additional reading about these topics, the teacher has designed an interactive task using a conversational agent: learners paste a teacher-written prompt into the tool and assign it a character living in Commonwealth London (a Protestant church leader, an anti-monarchist MP, a theatre manager, a university student or the mother of a bride) and then question that character about their background, daily routine, opinions and reactions to the political and religious changes of the period. The prompt instructs the system to remain historically plausible and consistent, to speak from the character's point of view with attention to occupation, age, marital status and political affiliation, to invent unspecified biographical details itself, and to give answers of interview length that invite follow-up. Learners may type or speak their questions and may receive answers as text or synthesised speech. In a second step, they write a short summary of five to seven sentences, without AI assistance, comparing what they learned against their prior expectations and citing concrete examples from the interviews.

Discussion

Safe use	Responsible use
The clearest compliance question in this sequence is the choice of the conversational agent. Some chatbots are built for school use, with teacher-created rooms that limit what learners must disclose; general-purpose AIs, on the other hand, often restrict use by minors and require individual accounts. For underage students this is a material difference, so this decision requires institutional checking rather than mere convenience. In addition to legal compliance, the teacher needs to be aware of some other **safety** requirements. For example, when designing the teacher-provided prompt, the teacher may want to consider how much information to disclose about the class, balancing pedagogical value with safety. In practical terms, this requires reflecting on whether identifiers, such as the learners' school year and curriculum are useful or could be removed without cost. Secondly, the offer of voice input to accommodate learning preferences introduces a category of data (speech recordings of minors) that warrants reflection.	The premise of the activity, namely that people in one city experience the same event in different terms, is squarely aligned with **Guideline 3**, and the design deserves credit for making perspective a structural feature. A critical analysis of the suggested cast can further improve the task, by adding absent voices: most of the suggested conversational participants (a church leader, an MP, a theatre manager, a university student) are literate, propertied male Londoners, and the one woman is defined by a marriage. A broader perspective might include, for example, the poor, the Jewish community readmitted in 1656, soldiers and servants. This requires reflecting on whether the model's fluency is adequate to the task of improvising all these characters without sounding identical. A related, easily missed issue is linguistic: the model will render 17th-century London in either contemporary standard English or a decorative archaism. Reflecting on which variety is being modelled, and why, converts an invisible default into a language lesson.

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	On **Guideline 4**, the ecological footprint of a handful of short conversations is modest, and the pedagogical case for dialogue over a jigsaw reading is defensible.
Purposeful use	Human-centred use
This is the sequence's strongest ground. Students must formulate questions in real time, interpret an unscripted answer, and adapt the next question accordingly, engaging in sustained, **meaning-focused interaction** that is genuinely difficult to arrange for a whole class at B1–B2. The requirement that the summary be written without AI assistance is a well-judged structural decision: it separates AI-mediated input from unaided production and keeps the assessable artefact the learner's own. In all, this is a genuinely transformative task, not because a chatbot has been deployed, but because it is the learners who set the line of inquiry and can run the same enquiry across several talking partners, an arrangement which would be impractical to stage without the affordances of AI. The pedagogical value of the task can be further sharpened by articulating clearly defined **linguistic objectives** (e.g., question forms, follow-up and clarification moves, reporting verbs, hedging of uncertain claims and the register of an interview). All of the above are directly implicated by the task and could be made explicit at little cost.	The fact that different students interviewing the same character type will meet different 'people' is a potential asset here: surfaced, it makes the **constructedness of the output** visible; unsurfaced, it simply produces contradictory notes. Given that the prompt instructs the model to invent plausible biographical detail and to fill gaps creatively, learners will receive confident, specific, unattributable claims and have no procedure for distinguishing them from evidence. A short third step would close the gap, by taking two or three concrete claims from a transcript back to the textbook or a source. On **Guideline 8**, the sequence generates an unusually good dataset for classroom-based inquiry: transcripts alongside unaided summaries. This enables the teacher to examine whether learners' questions become more historically specific across successive interviews, and which character types generate the richest talk.

Overall assessment

This case illustrates how a modest technological intervention can produce a substantive pedagogical shift when the design work is done well. Its strength lies not in the conversational agent itself but in the way the task distributes control: learners set the line of inquiry, sustain it across several talking partners, and are then required to account for what they found using their own words. The combination of learner-directed meaning-focused interaction and unaided production of written summaries makes the sequence transformative rather than merely efficient, and it exemplifies the kind of added value that Principle III envisages. The refinements suggested above are largely matters of making implicit decisions explicit: which tool, and on what institutional basis; which voices the cast includes and excludes, and how well the model can sustain them; which variety of English is being modelled; which language points the task is meant to develop; and how

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learners are to tell invented plausibility from historical evidence. None of these requires redesigning the lesson, and each converts a background assumption into an object of classroom attention. Read alongside the transcripts and summaries it generates, the sequence also offers the teacher a ready basis for inquiry into their own practice.

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Case 6: "Prepare next week for me": An agentic assistant plans a unit

Overview of the lesson

A teacher of Russian as a foreign language teaches an A1 class for adult learners, some of whom have studied some Russian at school and are better at reading and writing than they are in oral communication, and some of whom are heritage speakers who grew up speaking Russian at home but have relatively limited literacy skills. The teacher has been experimenting with an AI assistant that can (with the teacher's permission) read and write files in her cloud storage, access the gradebook and assignment functions of the learning management system (LMS), search the web, and send email from the teacher's account. She gives the assistant the following instruction: "*Prepare next week's unit on housing for my A1 class. Base it on last week's vocabulary quiz results, set up the homework in the LMS, and remind learners about Thursday's upcoming test.*" The assistant works through the task in a sequence of steps, without further input. One agent reads the quiz results and finds that six learners scored below 50%. Another one searches Russian-language property websites and copies five real flat-rental advertisements to use as authentic reading texts. A third agent drafts a four-lesson plan and creates worksheets at two levels. The assistant also then assigns the easier worksheet and an additional set of homework to the six low-scoring learners in the LMS, schedules the homework deadlines, drafts and sends the email. Since everything else in the unit is in Russian, it writes the email in Russian too. On Monday morning the teacher finds a summary of what was done and reads through the lesson plan, which looks well structured. The teacher does not check the individual steps.

Discussion

Safe use	Responsible use
This case differs from the ones discussed previously in that the main questions concern what the teacher delegated, rather than what the AI produced. Many actions that the agents carried out had consequences before any human saw them: for instance, homework was assigned to named learners, and an email went out under the teacher's name. **Guideline 1** warns against delegating tasks to agents that cannot be held legally, ethically or professionally responsible for the outcomes. Here, responsibility for the email and the assignments remains with the teacher, who did not review either. A second set of questions concerns permissions (**Guideline 2**). An assistant that can read the gradebook, write to the LMS and send email has access to a considerable amount of personal data. This raises questions about whether this level of access was approved by the	This case also illustrates how automated choices might unintentionally produce unfair outcomes (**Guideline 3**). Assigning remedial homework based on one signal (the test score), without considering the students' diverse backgrounds is unlikely to be neutral. This is a small-scale example of the risks of profiling discussed in Guideline 2. The email is the second example. Written in advanced Russian, it provided little usable information for many students. A teacher would not have made this mistake. The agent, working within the language of the unit, did not distinguish between materials for learners and communication with families. On **Guideline 4**, the environmental cost of agentic work is less visible than that of a single prompt, because one instruction can trigger many searches, generations and actions.

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school, and indeed whether the teacher knew what it included. Such decisions are best made at institutional level.	
Purposeful use	Human-centred use
The time that this teacher's workflow saves is considerable, and the lesson plan the AI produced is pedagogically sound. The use of authentic advertisements and differentiated worksheets aligns with both **Guidelines 5 and 6**. However, it is unclear whether teacher capacity has been expanded in this case, or simply bypassed. The plan follows a conventional sequence and draws on what is easy to find online, so it reflects typical practice rather than what this teacher knows about this class. A teacher who knows the heritage speakers might have used them as a resource in the unit, for example by asking them to explain how housing works in their families' experience. The agent made several decisions that are usually seen as the core of a teacher's professional judgment: which learners need support, what kind and how much. While delegating materials preparation to AI may be a defensible use, this is harder to justify when it comes to delegating decisions about individual learners without review.	The autonomy of agentic systems makes it harder to identify where a human exercised judgement. This case shows how it happens in practice. The teacher gave the instruction and read the summary, but the steps that mattered most happened between those two points. This is the gap that **Guideline 8** asks teachers to reflect on. Agentic tools usually let users set checkpoints (e.g., "show me the plan before producing the materials", "draft emails but don't send them"). Deciding where to place these checkpoints is a useful skill in AI-assisted lesson preparation. For the learners, the agent was invisible. The six who received extra exercises do not know that the decision was made by an algorithm, and so cannot question it.

Overall assessment

This case illustrates the potential of agentic AI when deployed in language education: in this case, it generated a full unit of work, including authentic materials and differentiated tasks, with minimal teacher involvement. It also demonstrates why the guidelines treat agentic systems as raising questions that other AI uses do not. The refinements suggested here focus on where human judgement needs to come in. The teacher should limit the agent's permissions to what the task needs, require approval before anything is assigned to learners or sent to families, and treat decisions about individual learners as the teacher's alone. Such refinements make the time-gains possible, while retaining teacher responsibility for the decisions they have to make.

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Glossary



Glossary

agency, for language learners and teachers, refers to their capacity to act purposefully and remain in control of learning and teaching practices, respectively. In the context of using AI for language education, the question of agency involves making informed choices about which aspects of language education are delegated to artificial intelligence and ensuring that learning experiences are consistently guided by human judgment and values.

agentic artificial intelligence is a type of AI that can pursue defined goals autonomously across multiple steps, taking actions and making decisions with minimal human intervention. In language education, agentic AI might be used to design and deliver sequences of learning activities, monitor learner progress, and adapt instruction accordingly.

artificial intelligence in language education refers to the ability of technology to undertake, with varying degrees of autonomy, assistive tasks that are normally subsumed into the roles of teachers, for the purposes of improving language learning. Such tasks include, for instance, lesson planning, materials preparation, providing opportunities for practice, and supporting assessment.

autonomy refers to the sense that one's actions and behaviours are self-determined as an authentic expression of the self and the self's internal values, even when these actions, behaviours and values may be influenced by external sources. According to Self-Determination Theory (SDT), only some intentional actions are truly self-regulated or autonomous (Ryan & Deci, 2017).

big data are datasets which cannot be processed or analysed using conventional data management tools, because of their volume, speed of generation, and variety (e.g., text, audio, video, or metadata). In language education, big data include – among other things – information about learner proficiency and achievement, personal profiles, and interaction logs from digital platforms, all of which might be used to generate personalised learning experiences or to inform educational policy.

classroom-based inquiry refers to a reflective and systematic set of practices that aim to identify how teaching can be improved, collecting and analysing data in the process of teaching, and using them to develop actionable insights. Classroom-based inquiry differs from formal research in that it prioritises teacher knowledge and because it is shaped by the particularities of the classroom under investigation (Kostoulas, 2026).

conversational agents (also known as chatbots or, outside language education, as assistants) are AI applications that simulate real-time dialogue with language learners. In language teaching and learning, conversational agents can help learners to practise speaking or writing in a target language, possibly also providing focused feedback on their output.

critical use of AI, in the context of these guidelines, refers to usages of AI that promote social justice. This definition of "criticality" is associated with: (1) an awareness of power structures, as opposed to treating AI as a 'neutral' tool, (2) a normative orientation invested in the expansion of democratic agency, (3) reflexivity, which involves awareness

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of the role of teachers and learners in the interaction with technology, and (4) a transformative intent. This definition differs from other uses of the word "critical", which foreground evaluation and fault-finding or higher-order cognition.

data mining is the automated analysis of large sets of learner input (e.g., learner profiles, voice recordings, writing samples) or metadata (e.g., engagement metrics, device information, location) in order to identify patterns. Such data mining can take place in order to personalise learning experiences, track progress, or generate learner profiles, but it can also be used for optimising applications or – more problematically – for commercial purposes.

feedforward is information provided to language learners to guide and improve their future output, for instance by offering guidance on how to apply suggestions or refine their language learning strategies. It is, in this sense, different from feedback, which typically focuses on what was done. In AI-supported language learning, feedforward might be derived from predictive analytics.

generative artificial intelligence is a type of AI that can generate new text or audio (or, less frequently in language education, images and code), from information in a large dataset. Some ways that generative AI applications have been used in language education include lesson plan generation, text adaptation, dialogue creation, or the production of language practice scenarios. Large language models, such as ChatGPT, are types of generative AI.

hallucinations, in the context of artificial intelligence, are instances where an AI system produces output that is fluent and plausible but factually inaccurate, or even completely fabricated.

intelligent tutoring systems (ITSs) are applications of AI that provide learners with personalised instruction and feedback, and adaptive opportunities for language practice. A strength of ITSs is that they analyse the learners' performance in real time, and can adjust the type, sequencing and difficulty of language learning tasks based on the learners' strengths, goals, and interests.

large language models (LLMs) are models trained on very large text corpora, which can be used to generate human-like language by predicting the most probable next word in a sequence. This makes them a subset of generative AI, with a focus on text. By analysing statistical patterns in language, LLMs can produce text that is coherent and contextually appropriate but is not based on actual 'understanding'. In language education, LLMs can be used to support conversational practice, scaffold writing, or generate learning materials.

metadata are information that describes or accompanies interaction with an AI application. Such information might include the time and location of an interaction, device type, browser settings, or engagement metrics (e.g., time spent on a task, click patterns).

predictive artificial intelligence is a type of artificial intelligence that looks for patterns in data to predict outcomes or suggest next steps. In language education, predictive AI

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can be used to help identify learners' strengths and weaknesses or to recommend personalised learning pathways.

profiling involves the use of learner responses or metadata to create personal learner profiles. Such profiles can be used to generate individualised learning pathways or adaptive feedback, but teachers and learners should be aware of possible risks such as reinforcing bias, narrowing opportunities, and the use of incomplete or context-insensitive data.

prompts are questions, instructions or similar input that teachers and learners provide to AI systems in order to elicit explanations, create texts, simulate conversations, or adapt materials.

reflexivity, in the context of language teaching, involves thinking about the connection between one's identity as a teacher (or teacher educator, or teacher researcher) and the activity in which they engage. According to Edge (2011), it requires reflection about two fundamental questions: (a) what difference does it make to my teaching (or teacher education, or teacher research) that it is *I, rather than anyone else*, conducting it? and (b) what difference does it make to me that I am conducting *this particular type* of teaching (or teacher education or teacher research)?

relatedness involves feeling socially connected. It is the sense of belonging and connectedness with others, which entails alignment with, and integration of, relevant social influences and values (Ryan & Deci, 2017).

Technological Pedagogical Content Knowledge, or TPACK, is a framework of knowledge described by Mishra and Koehler (2006), which positions effective technology integration as emerging from the intersection of three knowledge domains: (1) content knowledge, which encompasses subject matter, including linguistic, metalinguistic and cultural knowledge in the case of language education, (2) pedagogical knowledge, which involves broad knowledge of educational theory, learning processes, and curriculum design, and (3) technological knowledge, which involves understanding of and facility with digital tools and emerging technologies.

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Note on the use of artificial intelligence

Artificial intelligence (Claude Opus 5.5) was used in the preparation of this document to facilitate reference verification, consistent use of language across contributions from multiple authors, and adherence to ECML guidelines. The authors reviewed all AI output in all stages and bear responsibility for the content.

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